▶ Notebook

Lecture 5 at a glance: six parts, twenty-five sections

Part 1 · Why generate?

1.1 Things that do not exist ▶
1.2 Text, pictures and sound ▶
1.3 What p(x) means, and what it is for ▶
1.4 What they all share ▶

Part 2 · Autoencoders: squeeze, then rebuild

2.1 An image is 784 numbers ▶
2.2 Encoder, code, decoder ▶
2.3 The loss, and training ▶
2.4 What the code holds ▶

Part 3 · Can an autoencoder generate?

3.1 Decode random codes ▶
3.2 Where the codes are ▶
3.3 What we want instead ▶

Part 4 · From an autoencoder to a VAE: three fixes

4.1 Fix 1: choose where codes live ▶
4.2 Fix 2: a cloud for each image ▶
4.3 Fix 3: pull the clouds to N(0, I) ▶
4.4 The loss the fixes give ▶

Part 5 · Why this loss: Lecture 1's ELBO, one image at a time

5.1 Same problem as Lecture 1 ▶
5.2 A model with a hidden code ▶
5.3 Run the story backwards ▶
5.4 A guide q close to the posterior ▶
5.5 The loss, derived from the ELBO ▶
5.6 Both terms, live ▶
5.7 What the VAE learned ▶
5.8 The KL weight β ▶

Part 6 · Putting it together

6.1 The whole lecture in maths ▶
6.2 Wrap-up ▶

> Six parts, twenty-five numbered sections, about 94 minutes. The ▶ Notebook link on every slide opens its section in Colab.

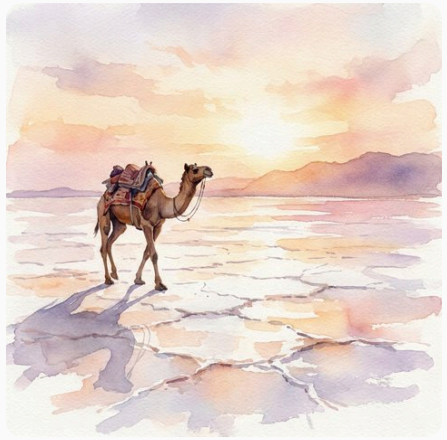
▶ Notebook

Part 1 · Why generate?

1 · Why generate? | 2 · Autoencoders | 3 · Can it generate? | 4 · Three fixes | 5 · Why: the ELBO | 6 · Putting it together



faces (StyleGAN)



text to image



a map to a landscape



digits: today

1.1 Things that do not exist ▶

1.2 Text, pictures and sound ▶

1.3 What p(x) means, and what it is for ▶

1.4 What they all share ▶

> Generative models make new text, pictures and sound. We take a quick tour, see what they are good for, and find the one recipe they all share.

▶ Notebook 1.1

Who are these people?



None of them exist. A network turned random numbers into each face.

> Nobody: each face is a sample from a generative model, StyleGAN (2019).

Faces generated by StyleGAN: Karras, Laine and Aila, "A Style-Based Generator Architecture for GANs", CVPR 2019. NVIDIA, CC BY-NC 4.0.

▶ Notebook 1.1

Real or generated? Six of these twelve digits are fake



Six are handwritten MNIST test digits. Six come from a network.

The fakes: numbers 2, 3, 5, 10, 11, 12, counting left to right, top to bottom.

By the end of today you will know how the network that made them was trained: a VAE with 16 numbers per code.

> The six outlined in red are generated. Some give themselves away with broken strokes; some fool most people.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 1.2

A chatbot is a generative model: one prompt, three different answers

Prompt: Write one sentence that opens a short story about the first monsoon rain in Gandhinagar.

Sample 1. As the first heavy drops of the monsoon struck the sunbaked avenues of Sector 21, the parched earth of Gandhinagar exhaled a sweet, dusty petrichor that signaled the end of the long summer.

Sample 2. As the first heavy drops struck the sun-baked asphalt of Sector 17, Gandhinagar exhaled its scorching summer heat in a sweet, petrichor-laced sigh that rippled down the quiet, tree-lined avenues.

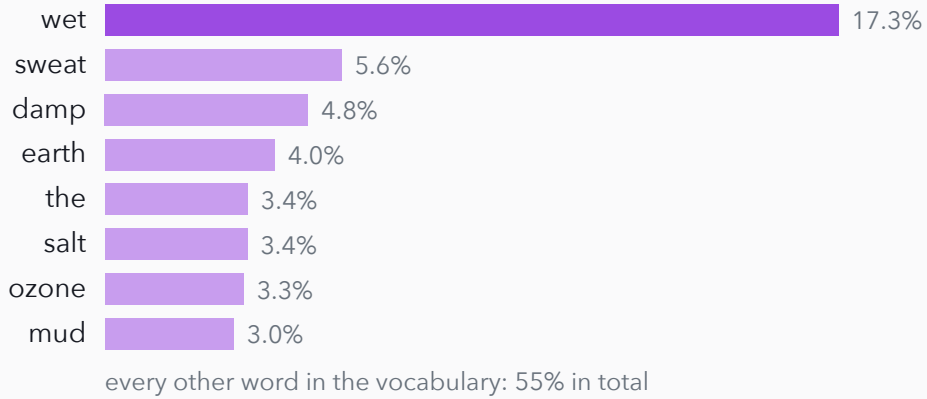
Sample 3. The brutal summer heat surrendered in a single afternoon when heavy, bruised clouds cracked open over Gandhinagar, washing its dusty neem trees and quiet, grid-lined sectors in the intoxicating scent of the season's first rain.

The same prompt gave three different replies. Each is a sample from a distribution over sentences.

▶ Notebook 1.2

Under the hood: a probability for every possible next word

“After months of heat, the first monsoon rain smelled of …”



Sampling = pick a word with these probabilities, append it, repeat.

The same model’s three continuations: “melons and oranges.”, “sweat and damp ruins. It was moist, it was s”, “desperation. What had occurred, and what transpired that its”.

> That loop is all a chatbot does when it writes, at a much larger scale.

SmolLM2-360M-Instruct (Hugging Face), next-token probabilities and three samples at temperature 1, computed for Lecture 2.

▶ Notebook 1.2

Text to image: one prompt, four samples

Prompt: A watercolour painting of a camel walking across the white salt desert of the Rann of Kutch at sunrise, soft light, wide view.



sample 1



sample 2



sample 3



sample 4

Three more runs of the same prompt bring a rider, a walker and a different horizon.

▶ Notebook 1.2

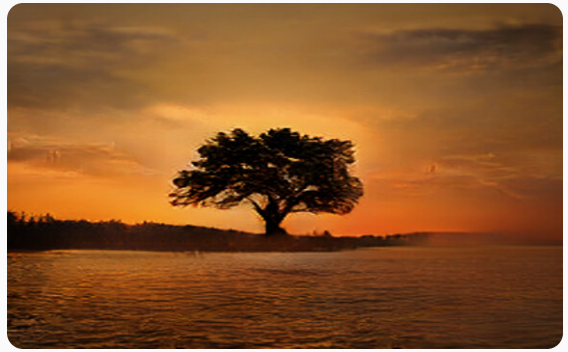
Image in, image out



one label map



sample 1



sample 2



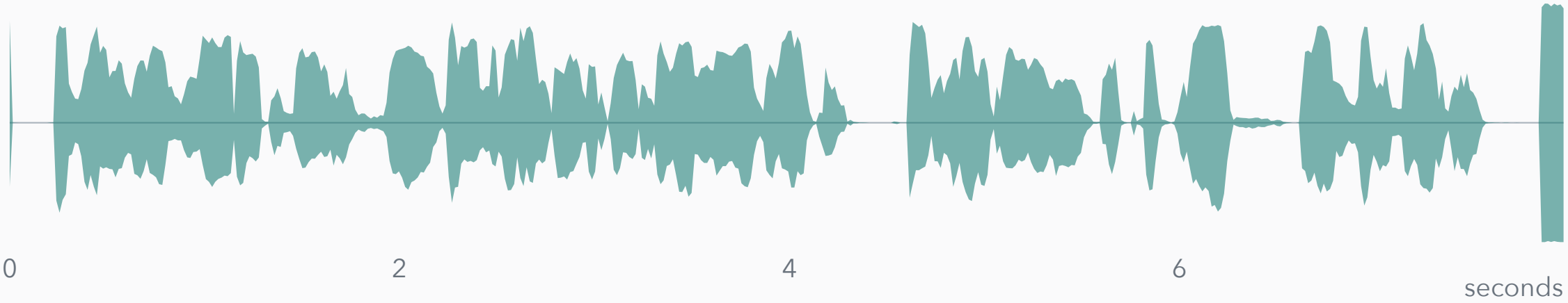
sample 3

> Sampling the same label map three times gives three different photographs. The model is still a distribution, now given an input.

▶ Notebook 1.2

Sound: text in, 24,000 numbers per second out

Text: “A generative model turns a few random numbers into a new example: a sentence, a picture, or this voice.”



0:00 / 0:00

191,219 numbers for 8.0 seconds. A few of them: 0.102, 0.089, 0.083, 0.072, 0.062, 0.051, …

The output is just a long list of numbers: the air pressure, 24,000 times a second. Generating sound means generating that list.

▶ Notebook 1.3

First, the names: x is the data, y is a label

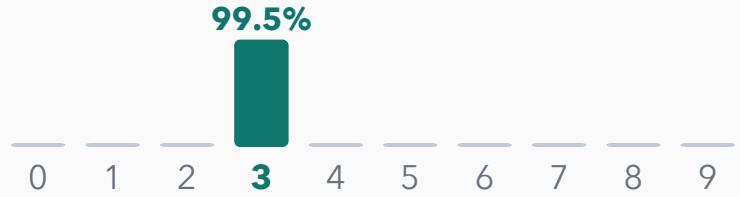
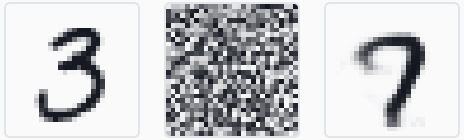
x = the data



one image: 784 numbers, one per pixel: (…, 0.00, 0.64, 0.99, 0.99, 0.99, …)

y = a label about x

3 which digit: one of 0, 1, …, 9

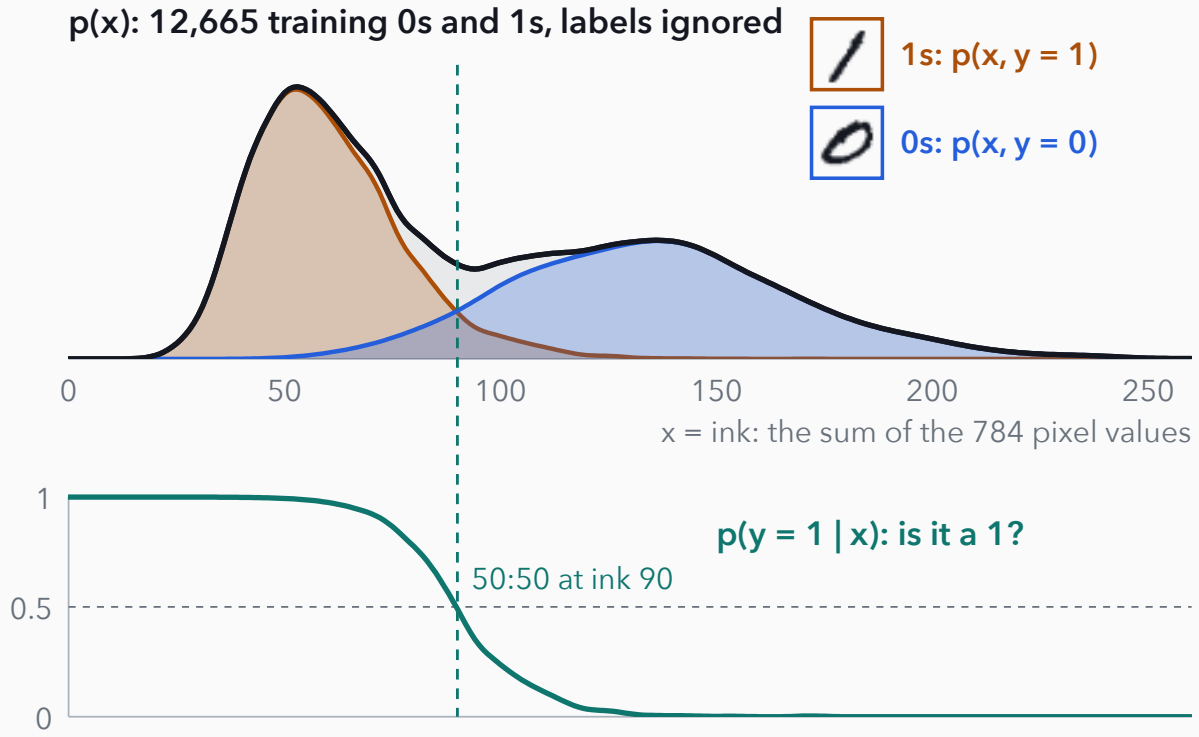
$p(y \mid x)$	Given this image, how probable is each label? "|" reads "given". **discriminative: a classifier**	
$p(x)$	How probable is this image itself, with no label? And how do we draw a new one? **generative: today**	
$p(x \mid y{=}3)$	Fix the label, then ask: which images? Every way to write a 3. **conditional generative**	

> p(x | y = 3) fixes the label first and covers every way to write a 3. The real 3s are draws from it. A text-to-image model learns p(image | prompt), with the prompt as y.

Classifier and numbers: `lecture5/scripts/xy.py`. Digits from Lecture 2's data; the new x is a draw from its 16-number VAE.

▶ Notebook 1.3

Generative or discriminative, in one picture



To draw it: one number per image, its ink, for the 0s and 1s only.

$$p(x, y) = p(y)\, p(x \mid y)$$

$$p(x) = p(x, 0) + p(x, 1)$$

$$p(y \mid x) = \frac{p(x, y)}{p(x)}$$

Generative learns the top: it can score and draw x, and the bottom follows.
Discriminative learns only the bottom: it can label, not draw.

Divide the 1s' hump by the total and you get p(y = 1 | x): near 1 for little ink, near 0 for a lot, 50:50 at ink 90. A classifier learns only this curve; a model of the humps gets it free by Bayes' rule.

Numbers: `lecture5/scripts/xy.py` (MNIST training set).

▶ Notebook 1.3

Beyond pictures: generative models in science

Weather

GenCast samples 50 possible futures for the next 15 days. The spread of the samples says how sure the forecast is.

Price et al., Nature 2025

Proteins

RFdiffusion generates new protein shapes, for example ones that bind a chosen target.

Watson et al., Nature 2023

Materials

MatterGen proposes new crystal structures with properties you ask for.

Zeni et al., Nature 2025

Missing data

Sensors drop readings. A model of p(x) fills the gap with plausible values instead of a flat average, and says how unsure it is.

for example, air-quality and smart-meter records

> The same goes for new materials, and for an everyday job: filling in missing sensor readings.

Price et al., "Probabilistic weather forecasting with machine learning" (GenCast), Nature 2025; Watson et al., "De novo design of protein structure and function with RFdiffusion", Nature 2023; Zeni et al., "A generative model for inorganic materials design" (MatterGen), Nature 2025.

▶ Notebook 1.3

What a model of p(x) is good for

Create

new text, images, sound, designs

Fill in

missing pixels, missing readings, the rest of a sentence

Simulate

many plausible futures instead of one average

Spot the odd one out

a reading with very low p(x) is suspicious: a faulty sensor, a fraud

Compress

likely data needs fewer bits: Lecture 1's codes

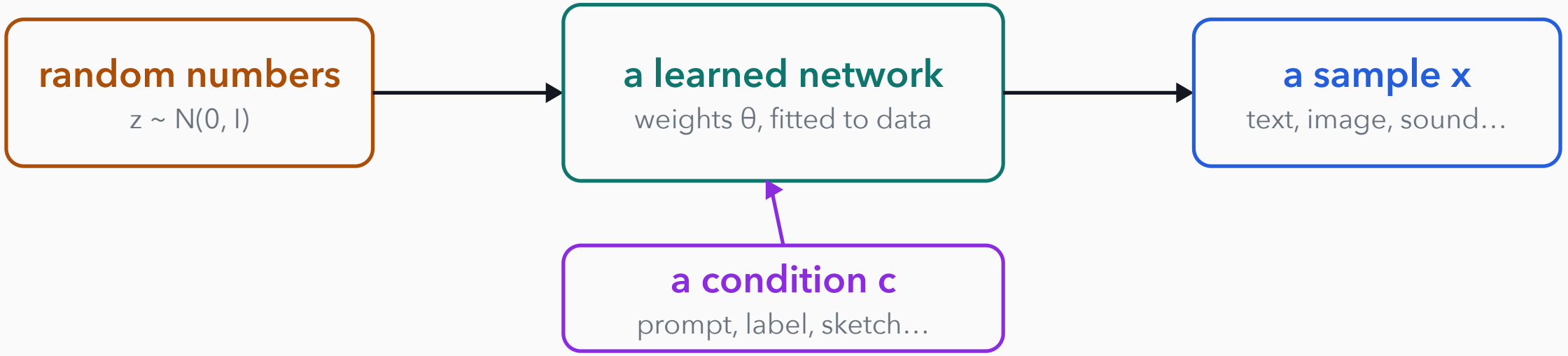
Make more data

extra training examples, or shareable synthetic records

> Spotting the odd one out and compressing need a score p(x). Making more data needs samples again, and helps when real data is scarce or private.

▶ Notebook 1.4

What they share: random numbers in, a learned network, a sample out



> Most also take a condition c that steers the sample: a prompt, a class label, a sketch. Today we drop c and learn p(x) for digits.

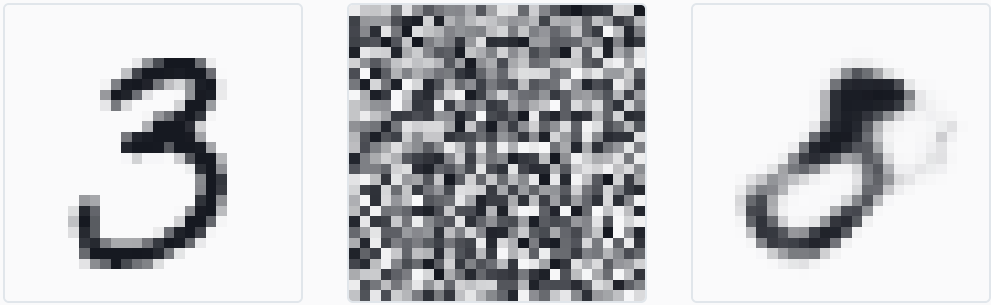
▶ Notebook 1.4

A model of p(x) can do two jobs: score and sample

Score: how plausible is this x?

$$p(\text{a clean } 3) \text{ high}, \qquad p(\text{static}) \text{ tiny}$$

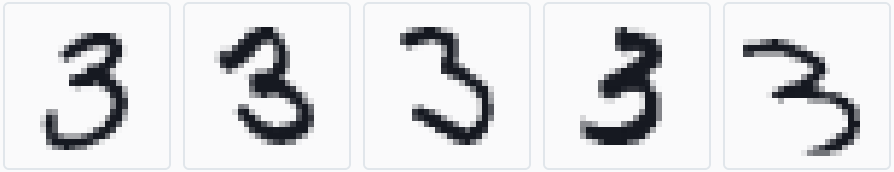
Sample: make a new x that looks like the data.



> Job two, sample: produce a new x. Both jobs come from the same fitted distribution.

▶ Notebook 1.4

Why not just output the average?



average of 100 threes:



Nobody writes this 3. The average is the blur in the middle of many real 3s.

> We need the whole distribution, so that we can draw one sharp 3 at a time.

▶ Notebook 1.4

Today: two networks, then Lecture 1's objective

Part 2 · Autoencoder

One network squeezes a digit into 2 numbers; a second rebuilds it.

Part 3 · Can it generate?

Feed the second network random numbers. What comes out, and why?

Part 4 · Three fixes

Codes from N(0, I), a cloud per image, a KL pull: a new loss.

Part 5 · Why this loss

A model with a hidden code: the loss is minus Lecture 1's ELBO.

Part 6 · Together

Every term of the loss traced back to KL and the ELBO.

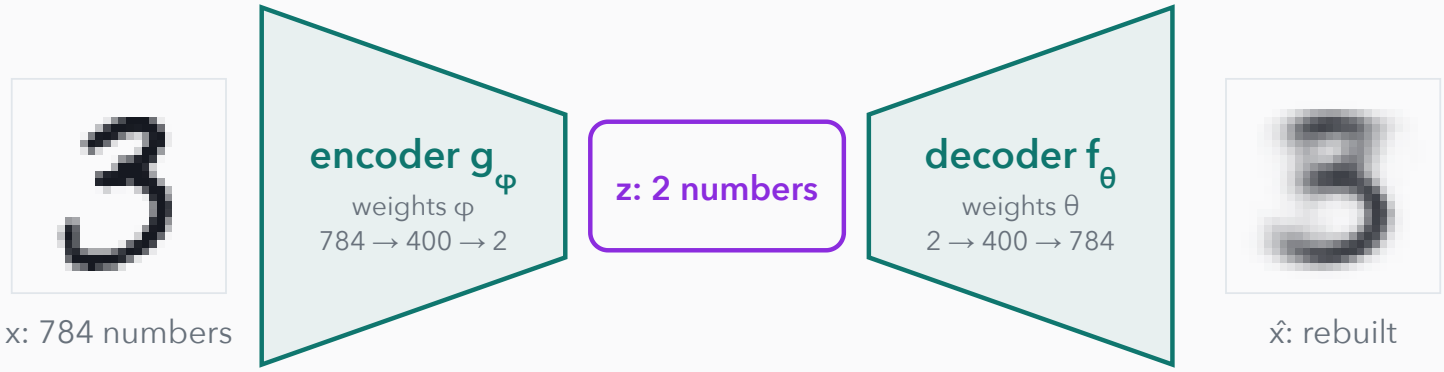
The VAE's loss for one image:

$$\underbrace{\frac{1}{2\sigma^2}\left\|x-\hat{x}\right\|^2}_{\text{rebuild it}}+\underbrace{\mathrm{KL}\left(q_\phi(z\mid x)\,\|\,\mathcal{N}(0,I)\right)}_{\text{Lecture 1's KL to the prior}}$$

> A last part traces every term of that loss back to KL and the ELBO.

▶ Notebook

Part 2 · Autoencoders: squeeze, then rebuild



2.1 An image is 784 numbers ▶
2.2 Encoder, code, decoder ▶
2.3 The loss, and training ▶
2.4 What the code holds ▶

> Squeeze each digit into 2 numbers, then rebuild it from them. Two small networks do this, trained together on one squared-error loss.

▶ Notebook 2.1

MNIST: 70,000 handwritten digits, 28 × 28 pixels each



0	0	0	0.3	0.8	1.0	1.0	1.0	1.0	1.0	1.0
0	0.0	0.5	1.0	1.0	0.9	0.6	0.3	0.2	0.5	1.0
0	0.4	1.0	0.9	0.6	0.1	0	0	0.0	0.5	1.0
0	0.1	0.6	0.1	0	0	0	0	0.2	1.0	1.0
0	0	0	0	0	0.0	0.4	0.6	0.9	1.0	0.9
0	0	0	0	0	0.3	1.0	1.0	1.0	0.9	0.2
0	0	0	0.6	0.9	1.0	1.0	1.0	1.0	1.0	0.4
0	0	0	0.9	1.0	1.0	1.0	0.7	1.0	1.0	1.0
0	0	0	0.0	0.2	0.0	0.0	0	0.2	0.7	1.0
0	0	0	0	0	0	0	0	0	0.1	0.9

60,000 for training, 10,000 for testing.

Our running example: test digit #90, a 3. Each pixel is a number: 0 is paper, 1 is full ink.

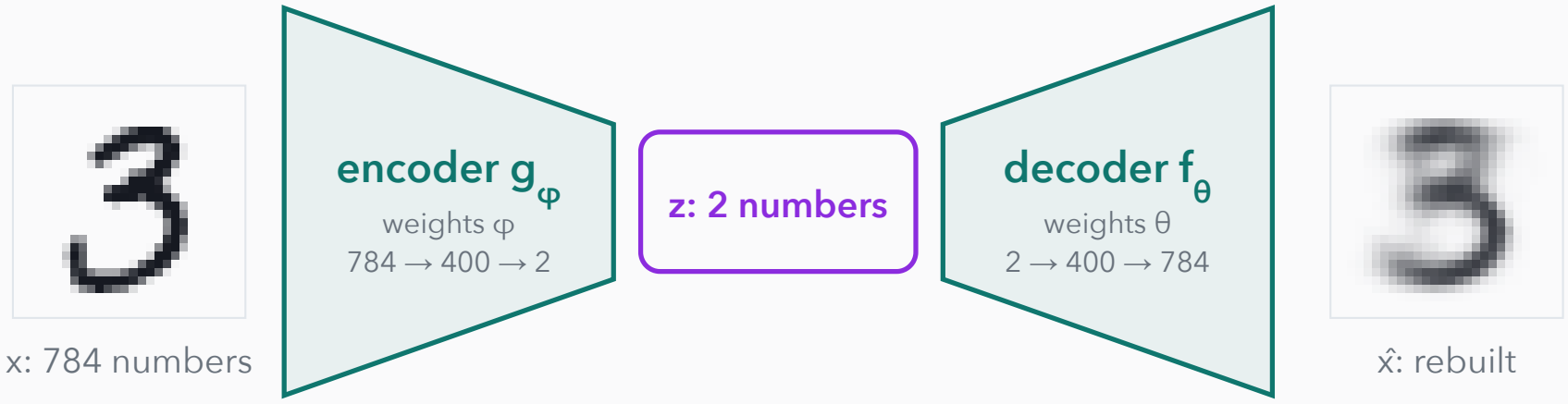
$$x \in [0, 1]^{784}$$

> We follow one digit all hour: test digit #90, a 3. To a network it is a list of 784 numbers between 0 and 1.

MNIST (LeCun, Cortes and Burges), grey levels scaled to [0, 1].

▶ Notebook 2.2

An autoencoder: encoder, short code, decoder



The decoder f_θ, with weights θ, mirrors it: 2 → 400 → 784, then a sigmoid. Train both so that $\hat{x} \approx x$.

▶ Notebook 2.2

Two small networks in PyTorch

```python
class AutoEncoder(nn.Module):
    def __init__(self, z=2):
        super().__init__()
        self.encoder = nn.Sequential(  # g_φ, weights φ
            nn.Linear(784, 400), nn.ReLU(),
            nn.Linear(400, z))
        self.decoder = nn.Sequential(  # f_θ, weights θ
            nn.Linear(z, 400), nn.ReLU(),
            nn.Linear(400, 784), nn.Sigmoid())

    def forward(self, x):
        z = self.encoder(x)     # 784 -> z numbers
        return self.decoder(z)  # z -> 784 numbers
```

Why a sigmoid at the end? Are we classifying?

No: it keeps every output pixel between 0 and 1, like the data.

> The final sigmoid squashes each of the 784 outputs into (0, 1), the range of our pixels.

▶ Notebook 2.3

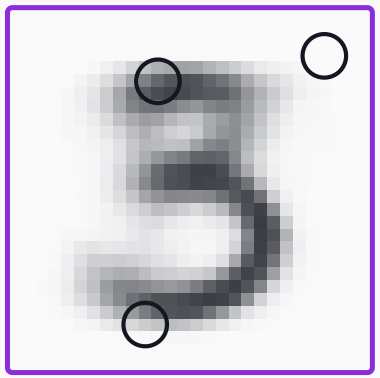
The loss: how far each rebuilt pixel is from the real one

$$L(\phi,\theta) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{784}\left(x_{ij} - \hat{x}_{ij}\right)^2, \qquad \hat{x}_i = f_\theta\left(g_\phi(x_i)\right)$$

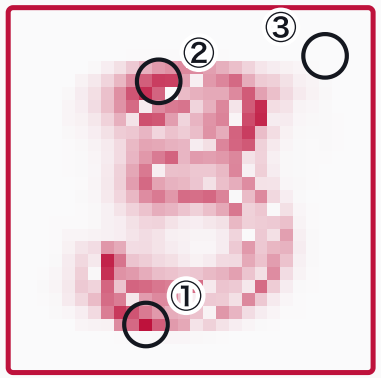
target x: our 3



prediction $\hat{x}$



$(x - \hat{x})^2$, each pixel



sum of all 784 = 26.3

pixel	x	$\hat{x}$	$(x - \hat{x})^2$
① missed ink	0.99	0.20	0.63
② extra ink	0.00	0.65	0.42
③ background	0.00	0.00	0.00
all 784 pixels			**26.3**

> Three pixels: ① missed ink costs 0.63, ② ink where the 3 has none costs 0.42, and ③ clean background costs 0.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 2.3

Before training: always look at what the network outputs



Every output pixel is between 0.38 and 0.66: flat grey, whatever the input.

Each pixel is off by about 0.5, since the target is 0 or 1. Squared: about 0.24.

$$\underbrace{784}_{\text{pixels}} \times \underbrace{0.235}_{\text{per pixel}} \approx 185$$

> Each of 784 pixels is off by about a half, which squares to a quarter: about 185 in all. Training must bring this far down.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 2.3

One training step: forward, loss, backward, update

```
model = AutoEncoder(z=2)
opt = torch.optim.Adam(model.parameters(), lr=1e-3)
for epoch in range(30):
    for x in loader:          # 128 images at a time
        x_hat = model(x)
        loss = ((x_hat - x)**2).sum(1).mean()
        opt.zero_grad()
        loss.backward()       # encoder AND decoder
        opt.step()
```

469 steps per pass over the 60,000 digits; 30 passes.

One loss, one backward pass: the gradient flows from the pixel errors back through the decoder, the code, and the encoder.

> loss.backward() reaches both networks, so the encoder and the decoder are trained together.

▶ Notebook 2.3

Press play: reconstructions sharpen, then stop improving

8 test digits



what it outputs after 469,000 steps (1,000 epochs)

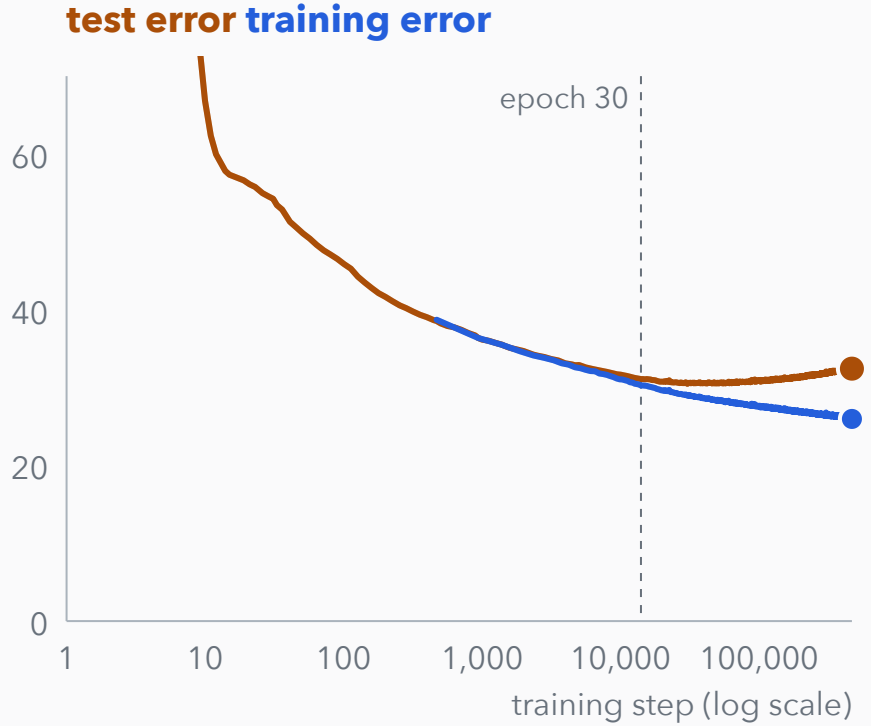


28 3 56 52 21 69 75 15

squared error of each digit (summed over its 784 pixels)

average over the 10,000 test digits: 32.4

average over the 60,000 training digits: 26.0



> Each digit takes shape: 31.1 after 30 epochs, where Lecture 2 stopped. Carry on to 1,000: the training error falls to 26.0, but the test error is lowest at epoch 65 (30.5) and ends at 32.4. Two numbers per digit, not training time, is the limit.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 2.4

More numbers in the code, sharper rebuilds

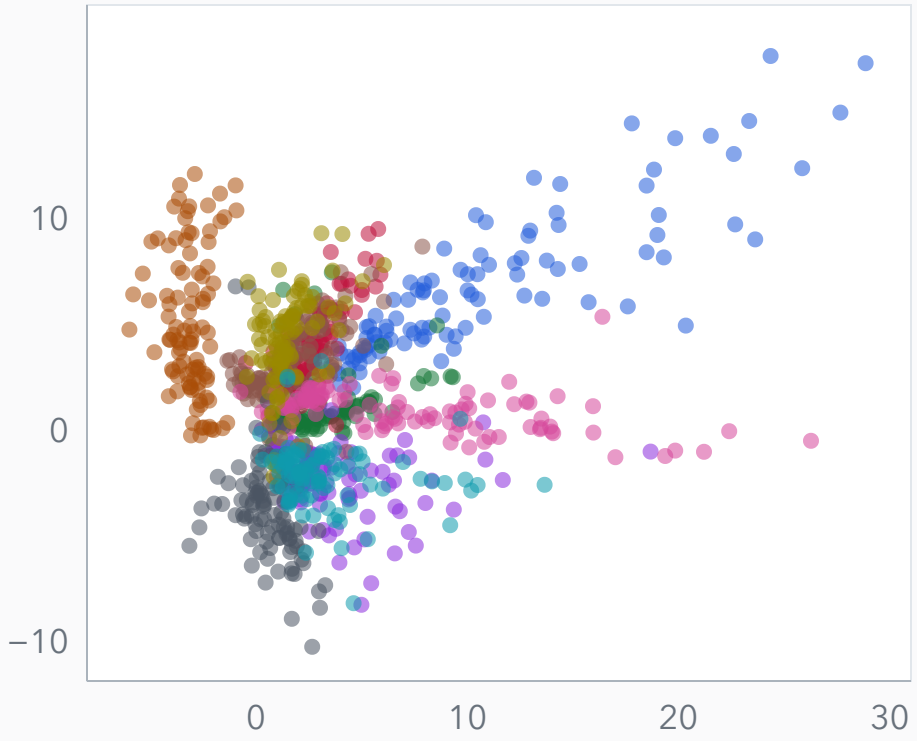


> With 32 and 128 numbers the strokes come back almost exactly. Part 3 asks which of these can generate.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 2.4

Where the codes land: 1,000 test digits at their 2 numbers



0 1 2 3 4 5 6 7 8 9

the whole code: z_1 and z_2

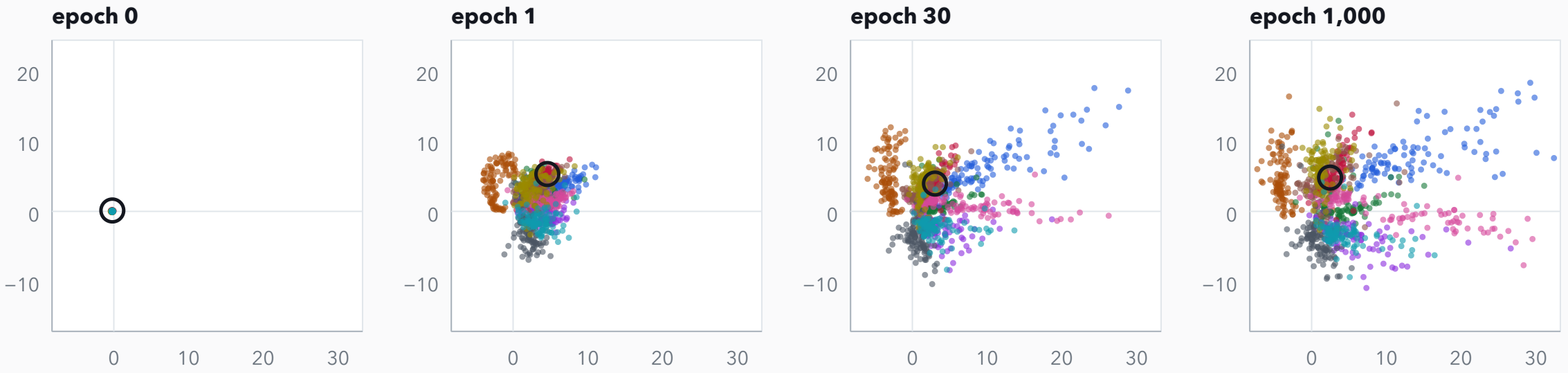
This autoencoder has exactly 2 numbers in its code, so nothing is hidden here. Hover any point to see its digit.

> Digits of one kind cluster together, although no labels were used in training. The codes run from about −5 to 22.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 2.4

During training: clusters form, then the codes drift out



train 184.5 · test 184.5

train 38.7 · test 38.5

train 30.3 · test 31.1

train 26.0 · test 32.4

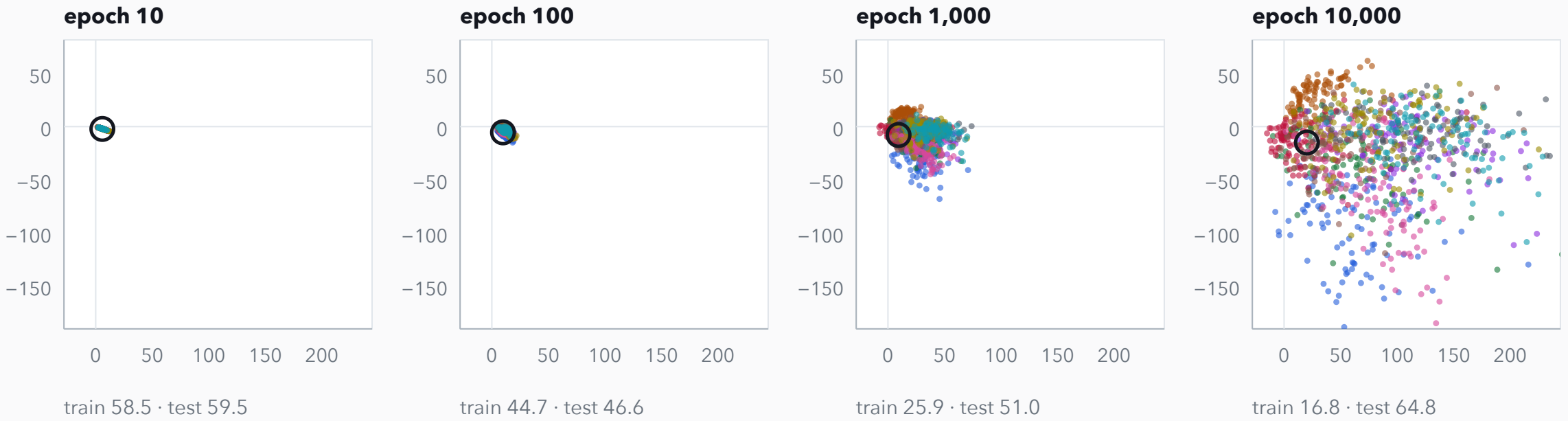
Trained on all 60,000 digits: the same axes in every panel. Our 3 is ringed.

> Press play. Epoch 30 is the last slide's picture; training goes on to 1,000. The clusters sharpen a little (71%) while the cloud keeps stretching: 13 wide after one epoch, 26 at 30, 33 at 300, 35 at 1,000. The training error keeps falling (26.0); the test error is lowest at epoch 65 and creeps up to 32.4.

Lecture 2's network and training, replayed and extended by `lecture5/scripts/epochs.py`.

▶ Notebook 2.4

Overfitting: too few digits, and the codes fly apart



train 58.5 · test 59.5

train 44.7 · test 46.6

train 25.9 · test 51.0

train 16.8 · test 64.8

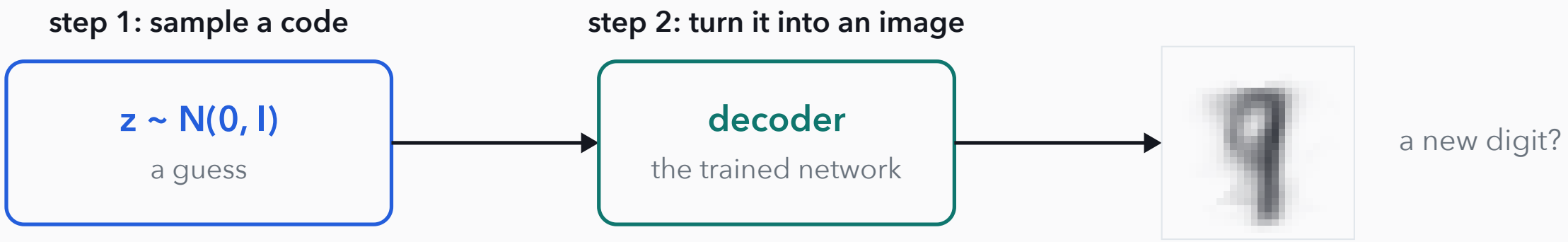
Trained on only 200 digits (20 of each): the same axes in every panel. Our 3 is ringed.

> Press play with “same axes throughout”: the codes fly outward, and the cloud grows from 14 wide at epoch 100 to 231. With so few digits the clusters never formed well (35% same-digit neighbours, against 70% before), and at the end our 3 comes back looking like one of the 200 training 2s.

`lecture5/scripts/epochs.py` (MNIST, fixed seeds).

▶ Notebook

Part 3 · Can an autoencoder generate?



3.1 Decode random codes ▶

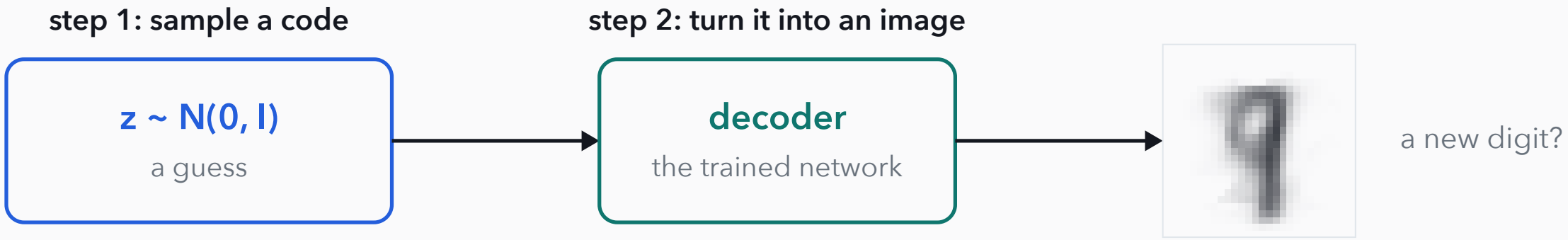
3.2 Where the codes are ▶

3.3 What we want instead ▶

> Keep the decoder, throw the encoder away, and invent the code. Does a random code give a digit?

▶ Notebook 3.1

Keep the decoder, feed it random numbers

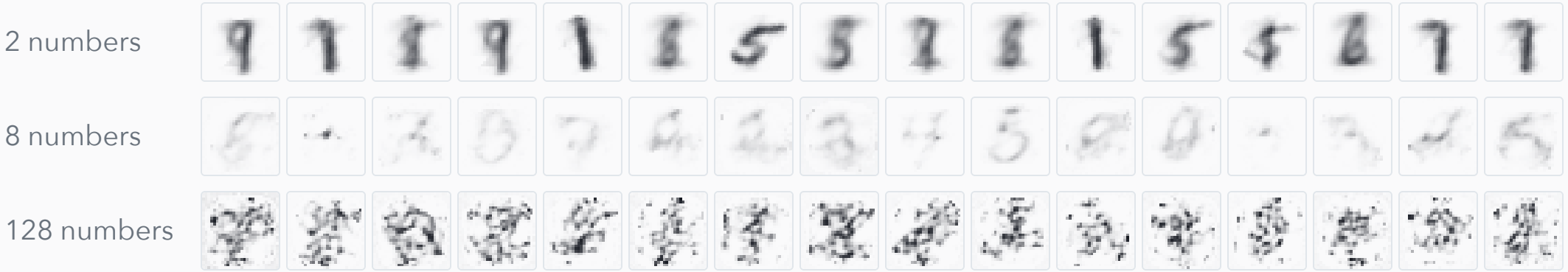


Which random numbers? The simplest: z ~ N(0, I), two independent standard bells. In code: `torch.randn(2)`.

> We need a distribution to draw codes from. Start with the simplest, the standard normal N(0, I).

▶ Notebook 3.1

Random codes through three autoencoders



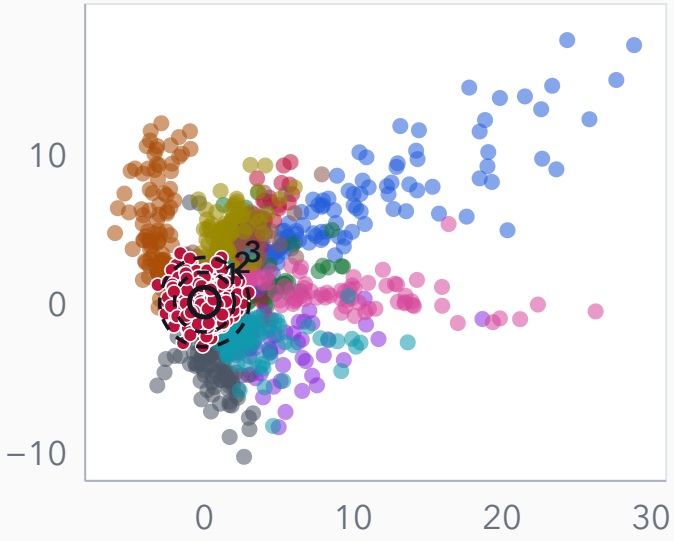
Takeaway. Better reconstruction did not give better samples: the best rebuilder makes static. Why? Look at where the codes are.

128 numbers, the best rebuilder of all, gives static.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Put the random codes on top of the real ones



0 1 2 3 4 5 6 7 8 9

real codes and 500 random ones (red)

Circles: N(0, I) at radius 1, 2 and 3.
Hover a dot to see the digit and what the decoder draws from that code.

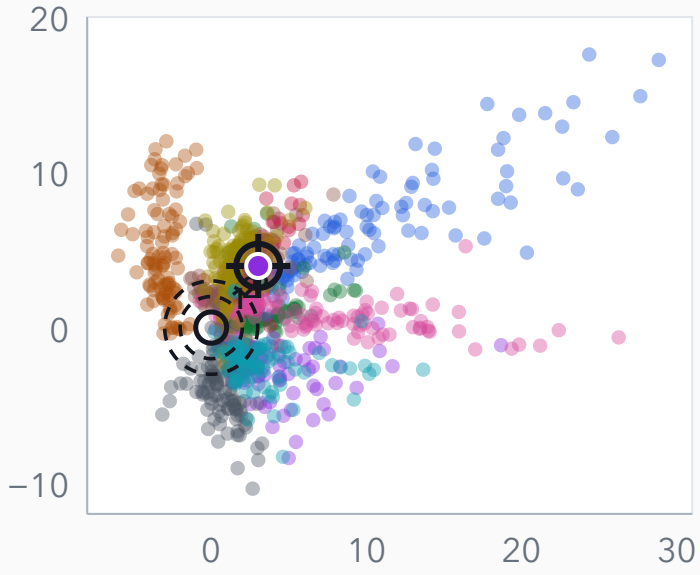
Takeaway. N(0, I) covers a small corner of the real codes: random codes reach a few digit types and miss the rest.

Nothing in the loss asked the codes to sit near N(0, I), so the red draws reach only the digits in one corner.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 3.2

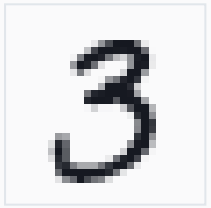
Your turn: three ways to choose a code



0 1 2 3 4 5 6 7 8 9

decoder of the autoencoder, run in your browser





circles: N(0, I)
at radius 1, 2, 3

the digit we encoded

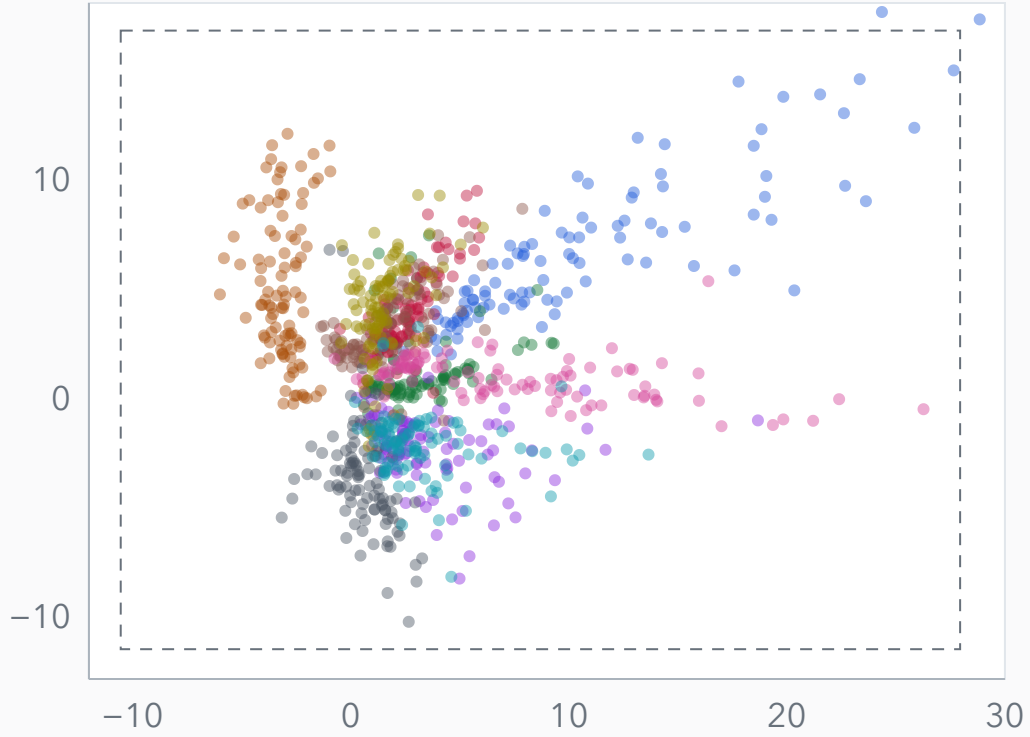
z = (3.02, 3.97)
the code of our 3, decoded

> Draw a random code from N(0, I), encode a real test digit, or click anywhere on the plane. The decoder runs live in the browser.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Decode a whole grid of codes





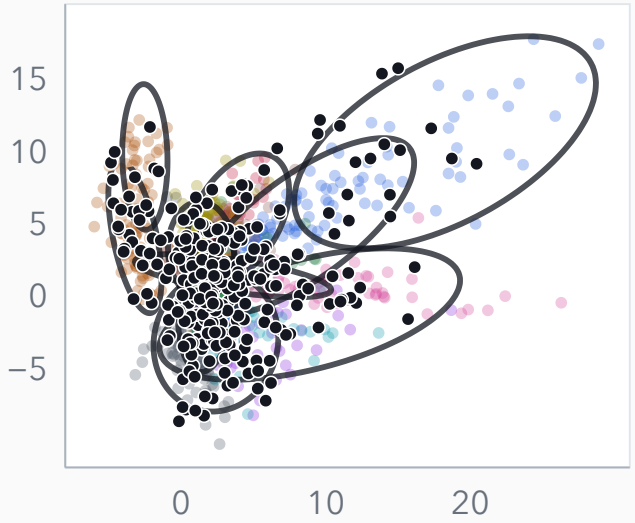
Takeaway. The decoder draws good digits only where real codes are. Away from them it extrapolates.

Over the real codes, digits blend into each other. Far from them, thick 1s and 0s and broken strokes: the decoder never trained there.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Why not sample from the real codes instead of N(0, I)?



1 · Copy: pick a training digit's code



2 · Jitter: a real code plus N(0, I) noise



3 · Fit 10 Gaussians to the codes, sample them



Takeaway. It works, and large systems do it: Stable Diffusion trains an autoencoder, then a second generative model for its codes. The VAE does it in one model: training puts the codes in N(0, I), and it can score p(x).

So a second model for the codes works. The VAE gets there with one model and one objective, and it also scores p(x).

Lecture 2's 2-number autoencoder and its 10-Gaussian fit to the training codes; draws and decodes: `lecture5/scripts/empirical.py`.

▶ Notebook 3.3

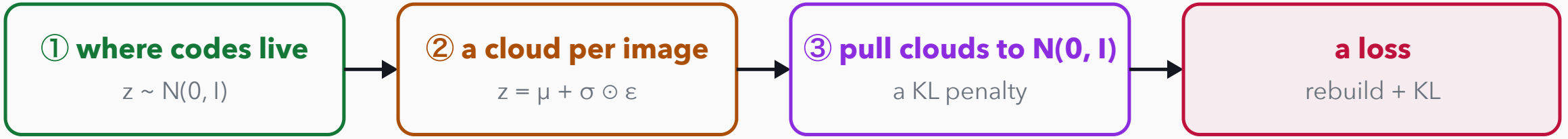
What we saw, and what we want instead

what we saw	what we want
random N(0, I) codes miss the real codes	codes that fill a region we know how to sample
gaps between codes decode to junk	a decoder trained everywhere in that region, gaps included
no way to say how likely an image is	one probability model that can score and sample

> Problem three: the autoencoder cannot score an image. We want a probability model. Part 4 fixes the first two; Part 5 shows the fix is that model.

▶ Notebook

Part 4 · From an autoencoder to a VAE: three fixes



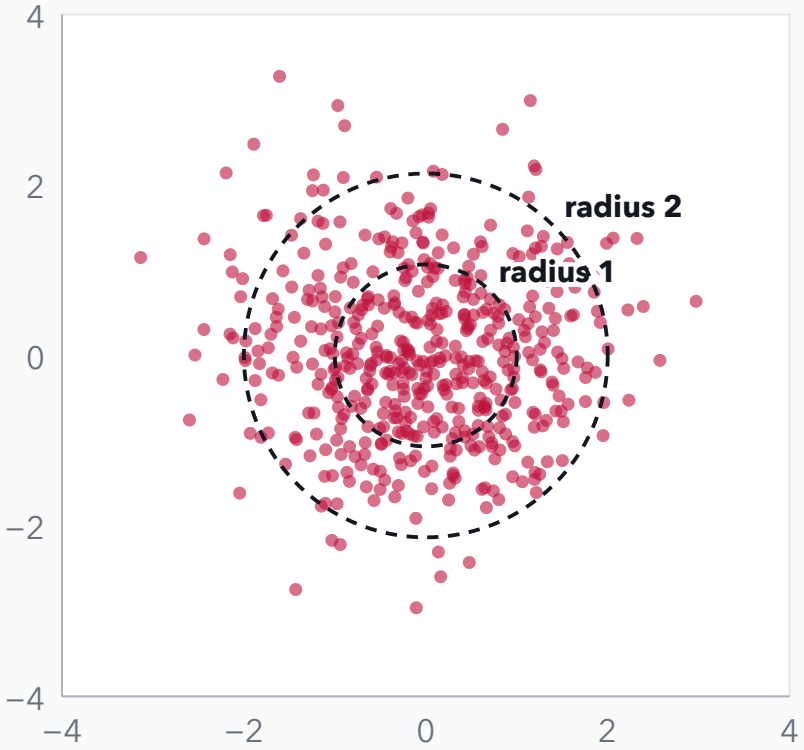
Part 5: this loss is exactly minus Lecture 1's ELBO, one image at a time

4.1 Fix 1: choose where codes live ▶
4.2 Fix 2: a cloud for each image ▶
4.3 Fix 3: pull the clouds to N(0, I) ▶
4.4 The loss the fixes give ▶

> The autoencoder cannot generate. Three fixes, one at a time, turn it into a VAE; each one answers a problem we saw in Part 3.

▶ Notebook 4.1

Fix 1: choose where the codes live, before training



Decide the distribution of codes in advance: z ~ N(0, I), two independent standard bells.

Why this one? It is easy to sample (`torch.randn(2)`), it has no gaps, and it is the same in every training run.

Generating becomes two lines: draw z, decode it. The work moves into training: the codes of real digits must fill this disc.

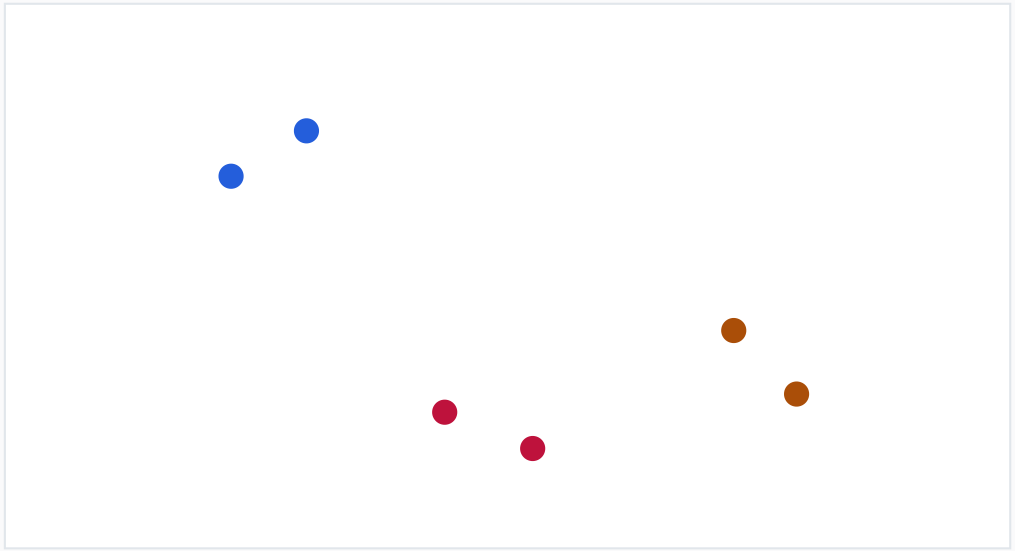
> If training makes the codes of real digits fill this disc, every random draw lands near some digit. Fixes 2 and 3 make that happen.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 4.2

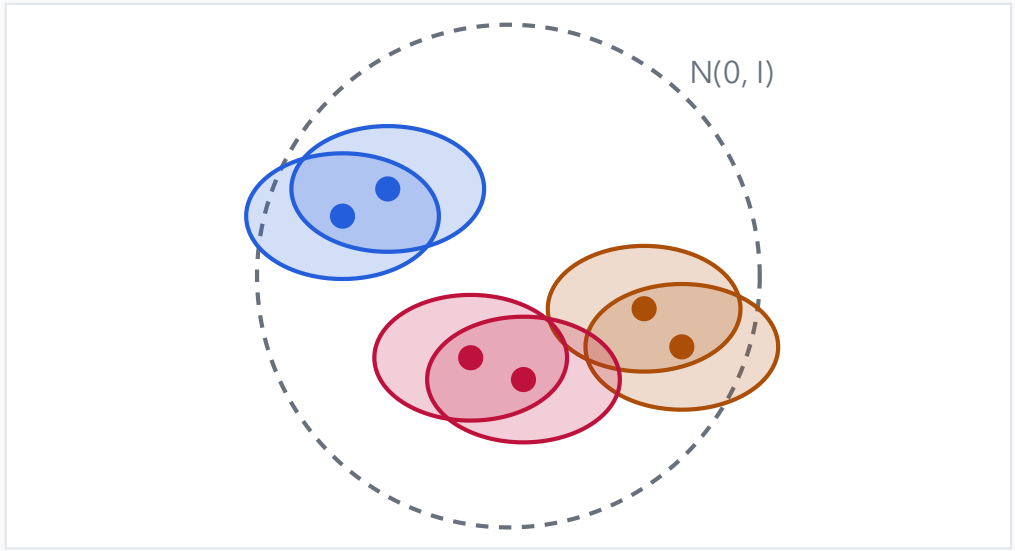
Fix 2: send each image to a small cloud of codes, not a point

autoencoder: each image → one point



points: the space between them is never trained

VAE: each image → a small cloud



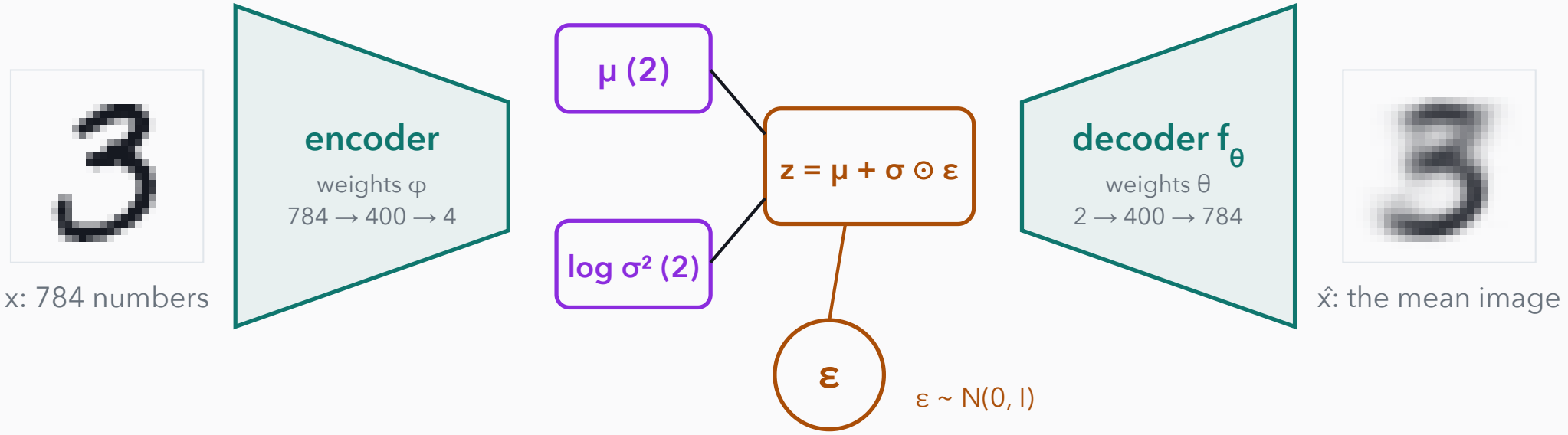
clouds overlap: the space between images is trained

A sketch, not data. Colours: three kinds of digit.

> Instead, send each image to a small cloud and decode a random point from it. Neighbouring clouds overlap, so the decoder learns the gaps too.

▶ Notebook 4.2

How a network outputs a cloud: a mean, a spread and a random draw



The decoder is the autoencoder's, unchanged: from z to a mean image.

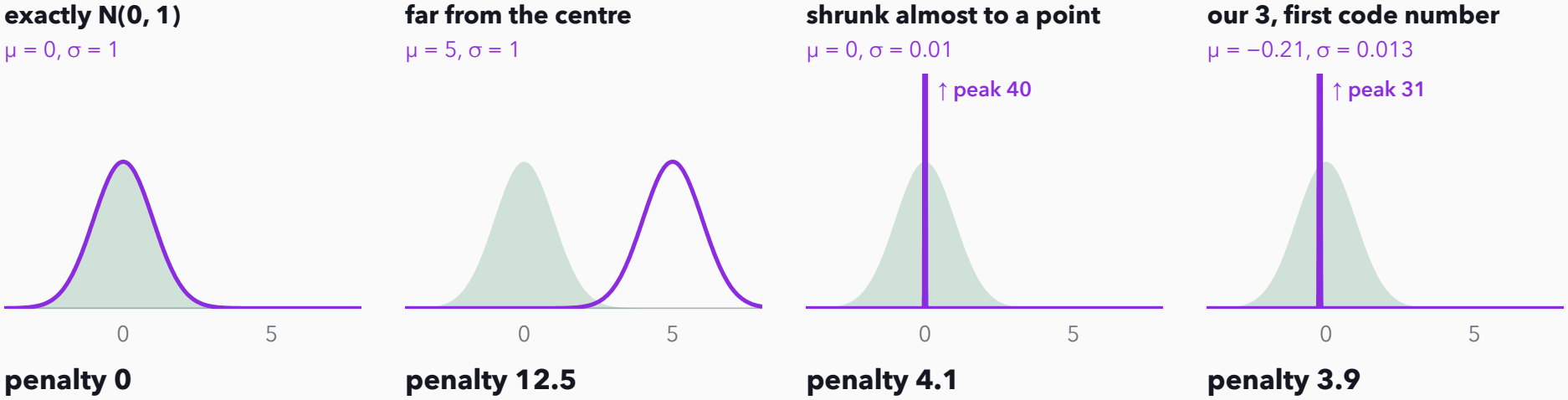
▶ Notebook 4.3

Fix 3: a penalty that pulls every cloud towards N(0, I)

With Fixes 1 and 2 alone, nothing stops the encoder from shrinking each cloud to a point and placing it far from the centre: an autoencoder again.

$$\text{penalty} = \mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, 1)\big)$$

↗ Lecture 1, 2.6: tool ②: the Gaussian KL



> Green is N(0, 1), purple the cloud. Far away costs 12.5; shrunk to a point costs 4.1, and our 3's cloud is nearly that narrow. Only N(0, 1) itself is free.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 4.4

The three fixes give a loss: reconstruction plus KL

fix	what it does	its formula	slide
Fix 1	choose where codes live	$z \sim \mathcal{N}(0, I)$	4.1
Fix 2	send each image to a cloud, and draw from it	$\mathcal{N}(\mu, \sigma^2), \quad z = \mu + \sigma \odot \epsilon$	4.2
Fix 3	pull each cloud towards N(0, I)	$\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, I)\big)$	4.3

$$L(x) = \underbrace{\big\|x - f_\theta(\mu + \sigma \odot \epsilon)\big\|^2}_{\textbf{reconstruction term}} + \underbrace{\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, I)\big)}_{\textbf{KL term}}$$

reconstruction term: rebuild x from a draw of its cloud (Fix 2)

KL term: keep the cloud close to $\mathcal{N}(0,I)$ (Fixes 1 and 3)

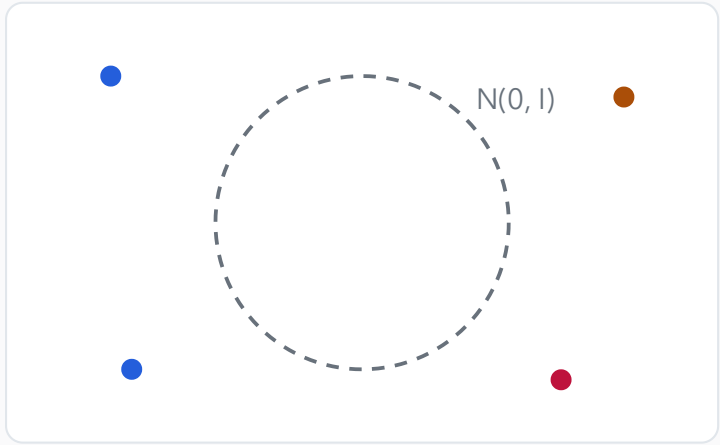
Two questions remain: how much weight each term should get, and whether this sum is a sensible objective or a hack that happens to work.

> Part 5 answers both with Lecture 1: the sum is minus the ELBO, the weight on the reconstruction term comes from the pixel noise, and the KL appears by itself.

▶ Notebook 4.4

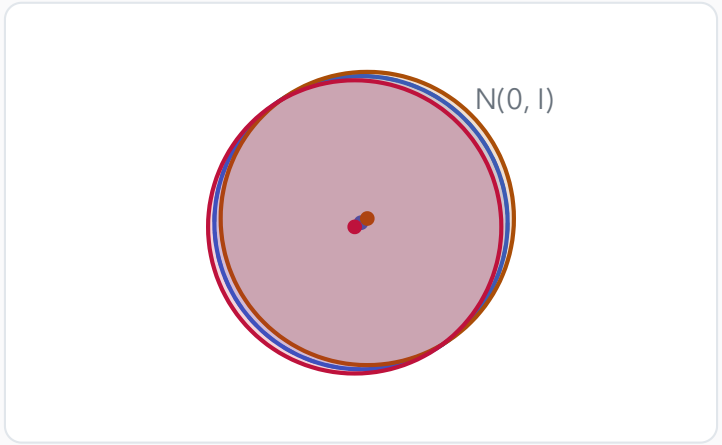
Two terms pull in opposite directions

reconstruction alone



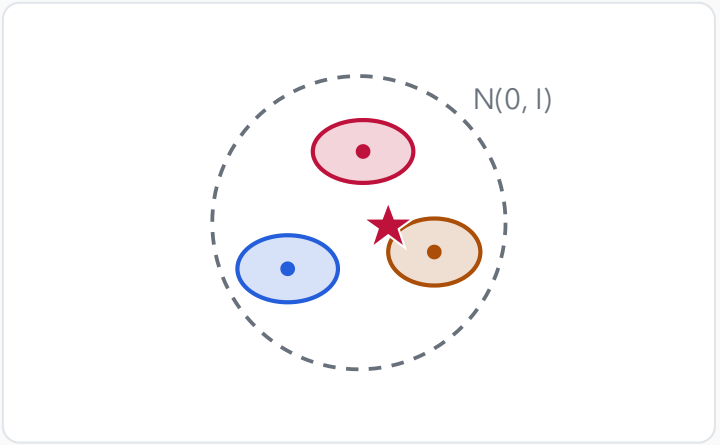
tiny clouds, far apart:
an autoencoder again

KL alone



every image gets the same cloud N(0, I):
z says nothing about the image

both together



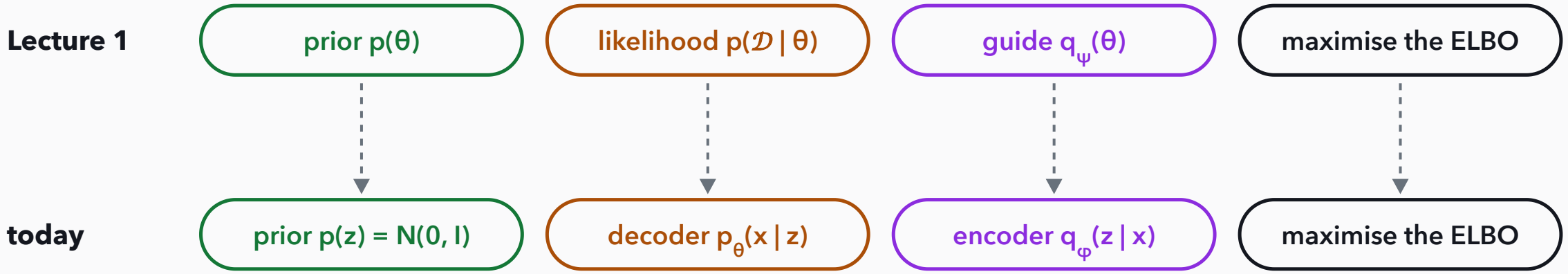
clouds near the centre, still apart:
a random z (star) lands near some cloud

Takeaway. A VAE is an autoencoder whose codes are noisy clouds held near N(0, I). To generate, draw z ~ N(0, I) and decode.

So a random draw from N(0, I) usually lands on or near some cloud, and the decoder has been trained to draw something there.

▶ Notebook

Part 5 · Why this loss: Lecture 1's ELBO, one image at a time



swap the names: the weights θ become a code z for each image, and the data $\mathcal{D}$ become one image x

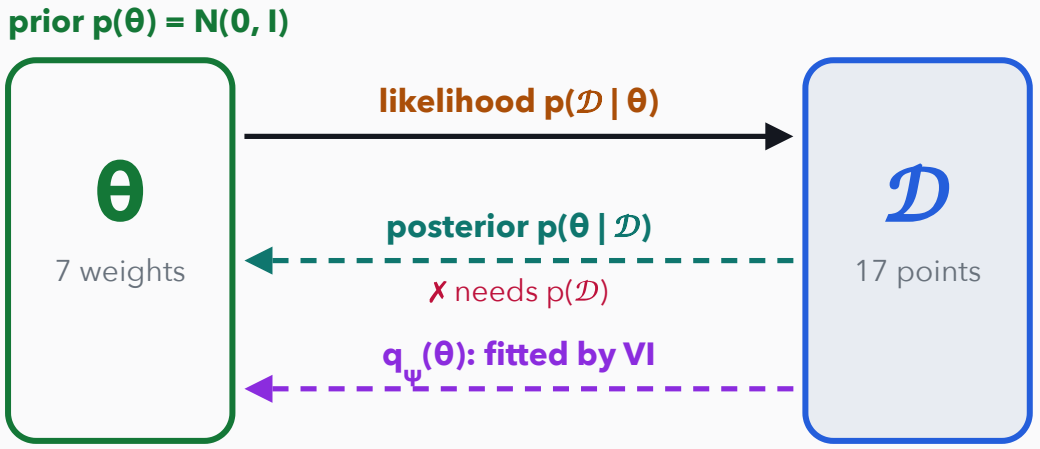
5.1 Same problem as Lecture 1 ▶
5.2 A model with a hidden code ▶
5.3 Run the story backwards ▶
5.4 A guide q close to the posterior ▶
5.5 The loss, derived from the ELBO ▶
5.6 Both terms, live ▶
5.7 What the VAE learned ▶
5.8 The KL weight β ▶

> Why is Part 4's loss the right one? Tell the model's story from a code, ask which codes made an image, hit Lecture 1's wall, then make a guide q close to the posterior, and the loss falls out.

▶ Notebook 5.1

Same problem as Lecture 1: the same picture, new names

Lecture 1: a tiny network

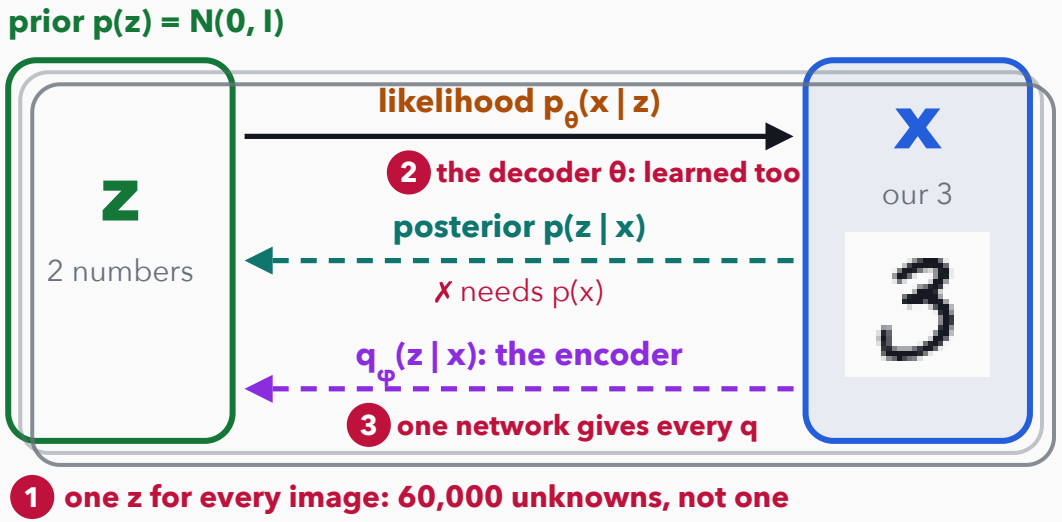


θ → z

𝒟 → x

same roles, same colours

Today: a VAE, for one image



$$\log p(\mathcal{D}) = \mathrm{ELBO}(q) + \mathrm{KL}\big(q \,\|\, p(\theta \mid \mathcal{D})\big)$$

$$\log p_\theta(x) = \mathrm{ELBO}(x) + \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z \mid x)\big)$$

> Three differences: ① every image has its own z, 60,000 unknowns; ② the likelihood has weights of its own, the decoder, learned at the same time; ③ one network, the encoder, gives every image its q.

▶ Notebook 5.1

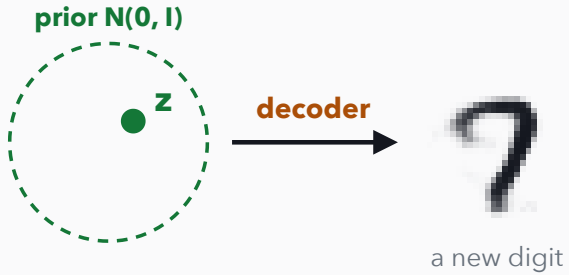
Why does a VAE need the posterior at all?

Lecture 1: the posterior gives predictions

$$p(y^* \mid x^*, \mathcal{D}) = \int p(y^* \mid x^*, \theta)\, p(\theta \mid \mathcal{D})\, d\theta$$

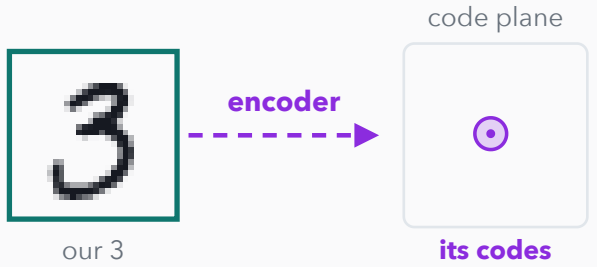
every network's prediction at a new x*, weighted by the posterior

A VAE generating: no posterior needed



a code from the prior, then the decoder

A VAE encoding: this is the posterior



which codes could have made this 3? That question is p(z | x).

Takeaway. A VAE is still an autoencoder. The decoder runs the story forwards, code → image; the encoder runs it backwards, image → codes. Running the story backwards is the posterior, so the encoder is our guess at it.

The decoder is the likelihood, and we can compute it. Training needs the posterior as well; Section 5.3 shows why, with our 3.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

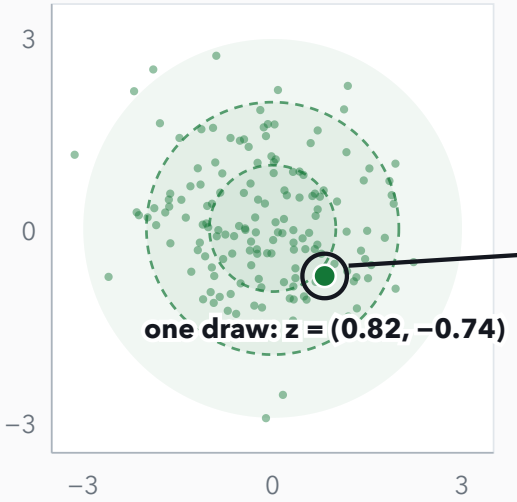
▶ Notebook 5.2

The model's story: first a code from the prior, then an image

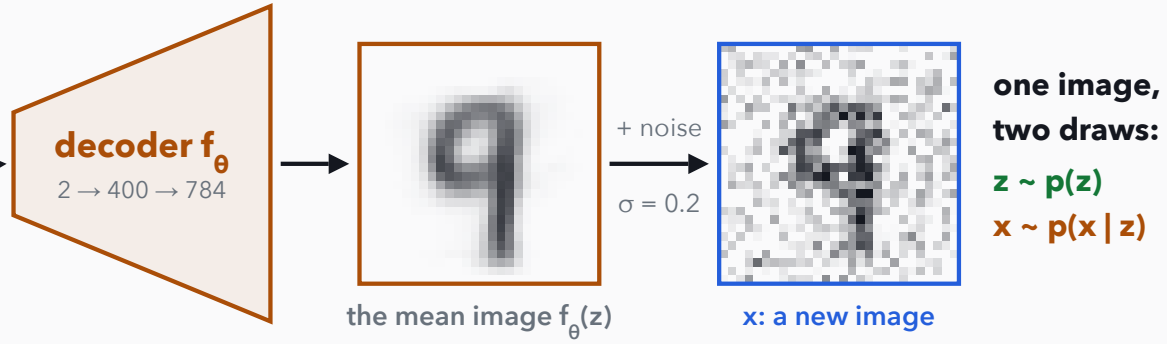
the model → score a code → which codes made x? → a guide q close to the posterior → the guide is a network → the loss

① the prior p(z) = N(0, I)

where codes come from, before any image



② the likelihood p(x | z): decode z, then add pixel noise



one image, two draws:

z ~ p(z)

x ~ p(x | z)

$$p_\theta(x, z) = p(z) \cdot p_\theta(x \mid z)$$

> Then every pixel gets its own noise of size σ = 0.2. Two draws make one image: z from the prior, then x from the likelihood. Their product is the joint p(x, z).

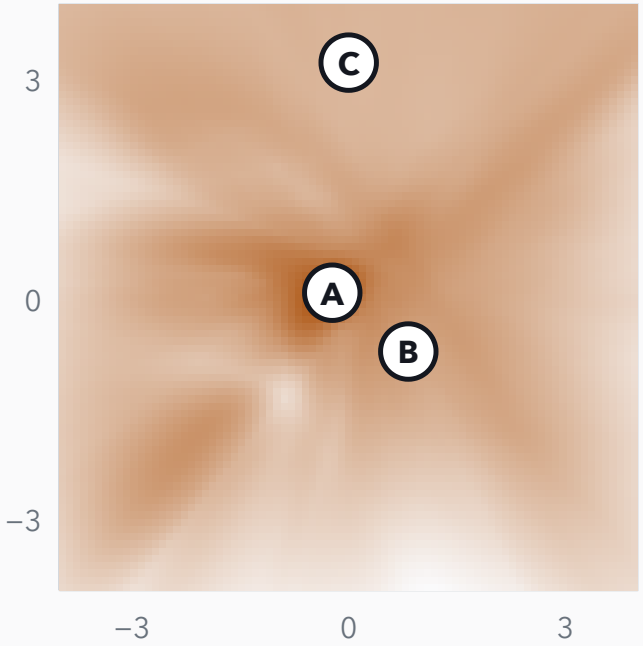
Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.2

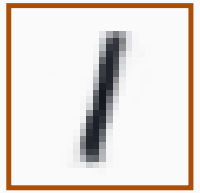
The likelihood scores a code: how well does it draw our 3?

the model → **score a code** → which codes made x? → a guide q close to the posterior → the guide is a network → the loss

the score of every code (darker: better)



best on the grid: 248, near A; worst: −992

code	what it draws		our 3	score: log p(x | z)
A		vs		**252** draws a 3: the best of the three z = (−0.23, 0.06)
B		vs		**−360** draws a 9: far lower z = (0.82, −0.74)
C		vs		**−505** draws a 1: lower still z = (0.00, 3.20)

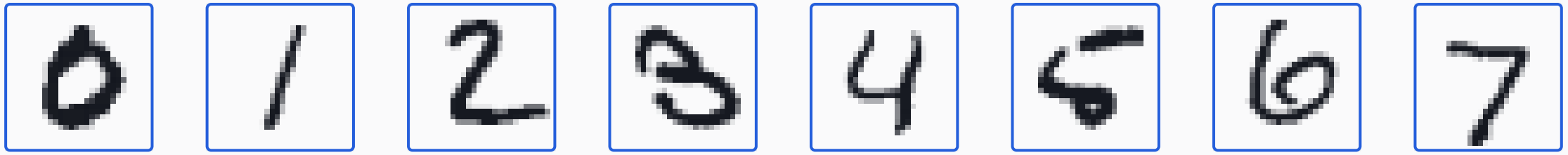
> Now score every code on the plane. Only a tiny patch near A draws our 3 well; almost everywhere else the decoder draws some other digit.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.3

Why run the story backwards? Images come without codes

the model → score a code → **which codes made x?** → a guide q close to the posterior → the guide is a network → the loss



z = ? z = ? z = ? z = ? z = ? z = ? z = ? z = ?

$$p_\theta(x) = \mathbb{E}_{z\sim p(z)}\left[p_\theta(x \mid z)\right]$$

how likely the model is to draw our 3 at all:
each code's chance of drawing it, averaged over the prior

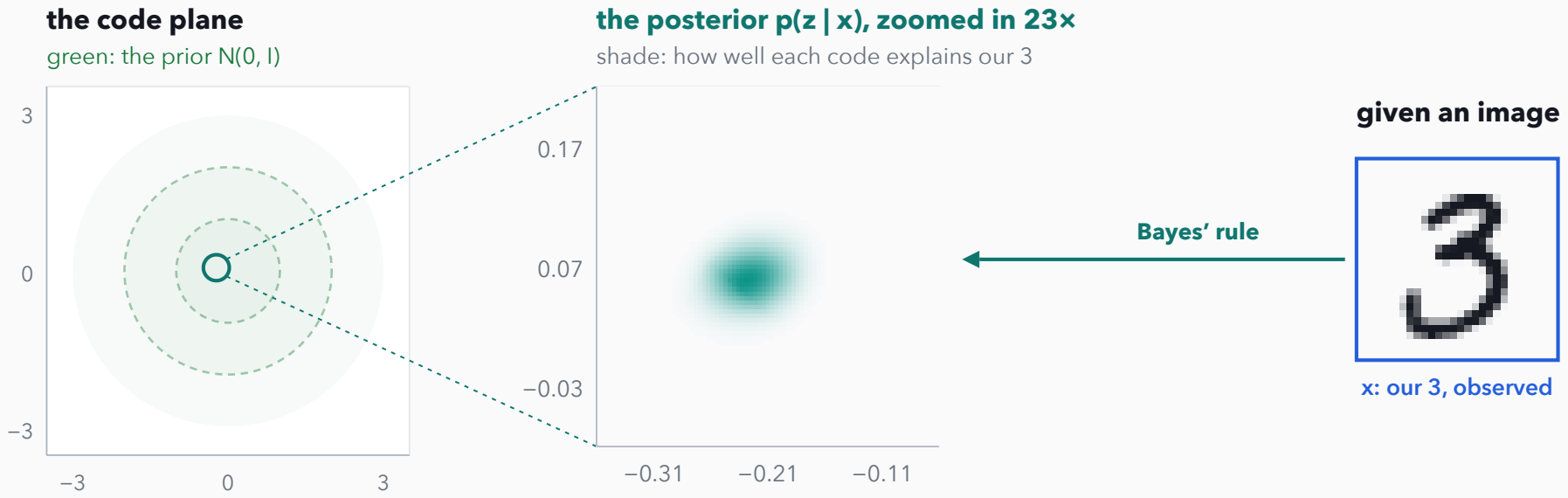
Takeaway. Codes like A carry this average; codes like B and C add almost nothing. So training needs to know which codes could have made our 3. That is the story run backwards, the posterior p(z | x).

To find the few codes that carry the average, run the story backwards. Lecture 1 asked the same of the weights θ: which ones explain the data?

▶ Notebook 5.3

Run the story backwards: which codes made our 3?

the model → score a code → **which codes made x?** → a guide q close to the posterior → the guide is a network → the loss



$$p(z \mid x) = \frac{p_\theta(x \mid z)\, p(z)}{p_\theta(x)} \qquad p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, dz$$

> The total we divide by is the evidence p(x): every code's likelihood, weighted by its prior. A grid adds it up here only because z has 2 numbers. Through a network in 16 numbers there is no formula: Lecture 1's wall.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.4

Variational inference: a guide q close to the posterior

the model → score a code → which codes made x? → **a guide q close to the posterior** → the guide is a network → the loss

the goal
a Gaussian guide q, close to the posterior

$$\min_q \ \mathrm{KL}\big(q(z) \,\|\, p(z \mid x)\big)$$

the KL, written out
an average over q

$$\mathrm{KL}\big(q \,\|\, p(z \mid x)\big) = \mathbb{E}_{z\sim q}\big[\log q(z) - \log p(z \mid x)\big]$$

Bayes' rule
the posterior, in logs

$$\log p(z \mid x) = \log p_\theta(x \mid z) + \log p(z) - \log p_\theta(x)$$

put it in
and regroup

$$\mathrm{KL}\big(q \,\|\, p(z \mid x)\big) = \underbrace{-\mathbb{E}_{z\sim q}\big[\log p_\theta(x \mid z)\big]}_{\text{reconstruction term}} + \underbrace{\mathrm{KL}\big(q \,\|\, p(z)\big)}_{\text{KL to the prior}} + \underbrace{\log p_\theta(x)}_{\text{the same for every } q}$$

so
log p(x) does not depend on q

$$\min_q \ \mathrm{KL}\big(q \,\|\, p(z \mid x)\big) \iff \min_q \ \underbrace{-\mathbb{E}_{z\sim q}\big[\log p_\theta(x \mid z)\big] + \mathrm{KL}\big(q \,\|\, p(z)\big)}_{\textbf{the VAE loss} \,=\, -\mathrm{ELBO}(q)}$$

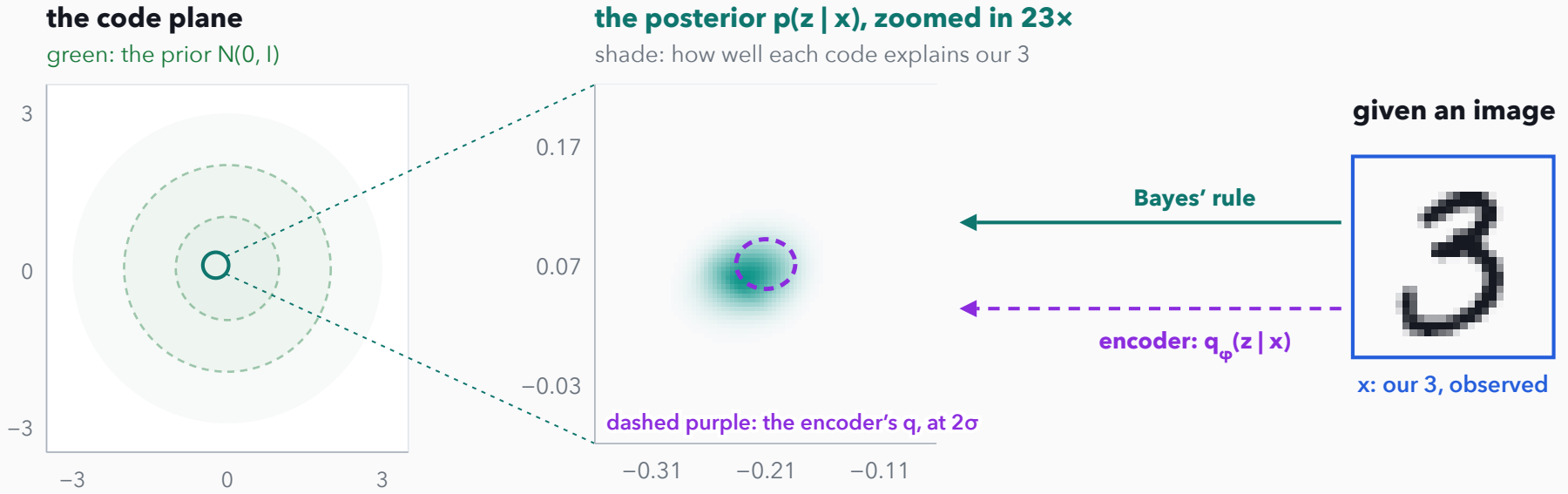
> log p(x) is the same number whichever q we pick, so dropping it does not change which q is best. What is left is the VAE loss, minus Lecture 1's ELBO.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.4

The guide is a neural network: the encoder

the model → score a code → which codes made x? → a guide q close to the posterior → **the guide is a network** → the loss



$$q_\phi(z \mid x) = \mathcal{N}\big(\mu_\phi(x),\ \sigma_\phi(x)^2\big)$$

The decoder tells the story, z → x.
The encoder, the guide, runs it backwards, x → z.

> Now we have two networks: the decoder p_θ(x | z) and the guide q_φ(z | x). Train both together on the last slide's loss, summed over all images.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.5

Put in the model's choices: the loss is reconstruction + KL

the model → score a code → which codes made x? → a guide q close to the posterior → the guide is a network → **the loss**

start
the loss from variational inference, with the encoder as the guide

choice 1
pixels: Gaussian around the decoder's output, noise 0.2

choice 2
prior N(0, I); q from the encoder: N(μ, σ²)

result
put both in: Part 4's loss

$$-\mathrm{ELBO}(x) = -\mathbb{E}_{z\sim q_\phi(z|x)}\big[\log p_\theta(x \mid z)\big] + \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big)$$

$$\log p_\theta(x \mid z) = -\frac{\|x - f_\theta(z)\|^2}{2 \cdot 0.2^2} + \text{const}$$

$$\mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big) = \mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, I)\big)$$

$$-\mathrm{ELBO}(x) = \underbrace{\mathbb{E}_q\left[\tfrac{1}{2\cdot 0.2^2}\|x - f_\theta(z)\|^2\right]}_{\text{reconstruction term: Fix 2}} + \underbrace{\mathrm{KL}\big(q_\phi \,\|\, \mathcal{N}(0, I)\big)}_{\text{KL term: Fixes 1, 3}} + \text{const}$$

> Put both in: the reconstruction term plus the KL term, Part 4's loss term for term, with its weight now decided. The constant does not change any gradient, so we drop it.

▶ Notebook 5.5

One step for our 3, with numbers

$$\mu = (-0.211,\ 0.070), \quad \sigma = (0.013,\ 0.010), \quad \epsilon = (0.5,\ -1.0)$$

$$z = \mu + \sigma \odot \epsilon = (-0.204,\ 0.060)$$



vs



$$\frac{1}{2(0.2)^2} \times 23.18 = 289.8$$

$$\mathrm{KL} = 7.94 \qquad L = 289.8 + 7.94 = 297.7$$

Add the KL to the prior, from tool ②: the loss for this image on this step.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.5

The VAE in PyTorch: three changes to the autoencoder

```
class VAE(AutoEncoder):                # same decoder
    def __init__(self, z=2):
        super().__init__(z)
        self.encoder[-1] = nn.Linear(400, 2 * z)  # (1)

    def forward(self, x):
        mu, logvar = self.encoder(x).chunk(2, dim=1)
        eps = torch.randn_like(mu)
        z = mu + torch.exp(0.5 * logvar) * eps       # (2)
        return self.decoder(z), mu, logvar

x_hat, mu, logvar = model(x)
recon = ((x_hat - x)**2).sum(1) / (2 * 0.2**2)
kl = 0.5 * (mu**2 + logvar.exp() - logvar - 1).sum(1)  # (3)
loss = (recon + kl).mean()
```

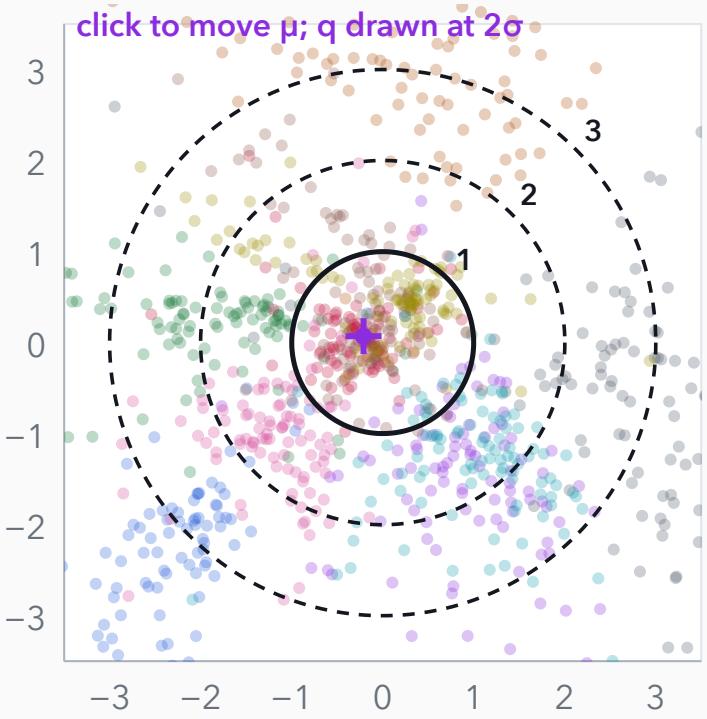
(1) the encoder outputs 2z numbers, (2) the code is a reparameterised draw (tool ①), (3) the loss adds the KL (tool ②).

The training loop is the autoencoder's, unchanged.

> Everything else, the batches, Adam and the passes over the data, is the autoencoder's loop.

▶ Notebook 5.6

Your turn: choose q for our 3, and watch both terms



x: our 3 **f(μ): decoded** five draws z from q, decoded



reconstruction	**290.1**	E‖x − f(z)‖² / 2σ²: small if draws rebuild our 3
KL to the prior	**7.9**	KL(q ‖ N(0, I)): small if q sits near N(0, I)
ELBO	**243.3**	= 541.4 − reconstruction − KL

gap to log p(x): 0.7, the KL from q to the posterior

log p(x) = 244.1: the ceiling

purple dot: the ELBO · red: the gap

−600 −300 0 300

q = the encoder's q for our 3: μ = (−0.21, 0.07), σ = 0.012.

> Try q = N(0, I), then q on the 1s. Every q stays below the ceiling log p(x); the red gap is the KL to the posterior.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.7

Now sample: z ~ N(0, I), then decode

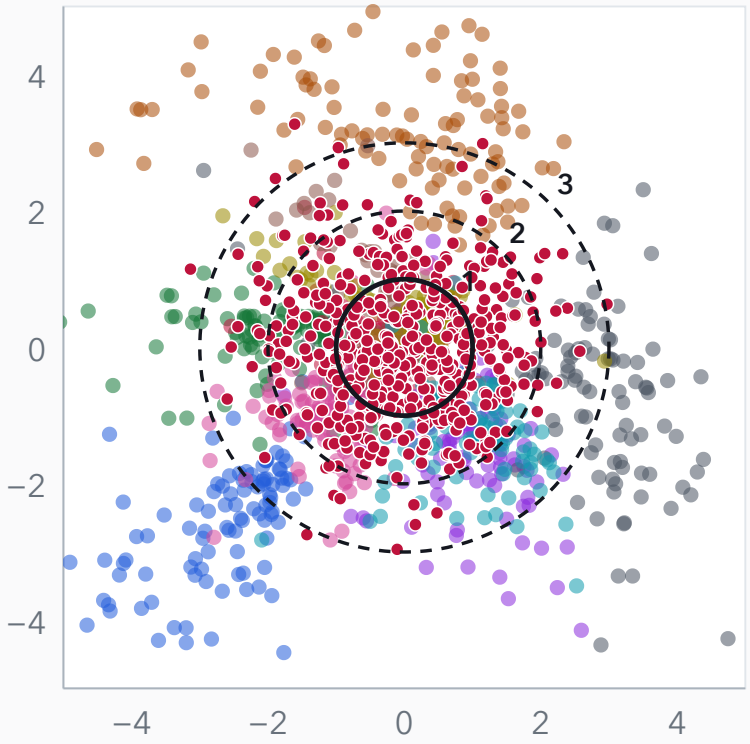


> Through the 16-number VAE: sharper and more varied. The fakes at the start of the hour came from this model.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.7

The codes now fill N(0, I)



Codes of all 10,000 test digits: mean (−0.2, −0.2), spread (1.8, 1.7).

Random draws (red) now land among real codes, on every digit.

> Overlay 500 draws from N(0, I). Because they land among real codes, sampling now works.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.7

Decode a grid of codes: digits everywhere

VAE: codes from −3 to 3 in each direction



walk in a straight line from a 1 to a 0



the same walk, 16-number VAE

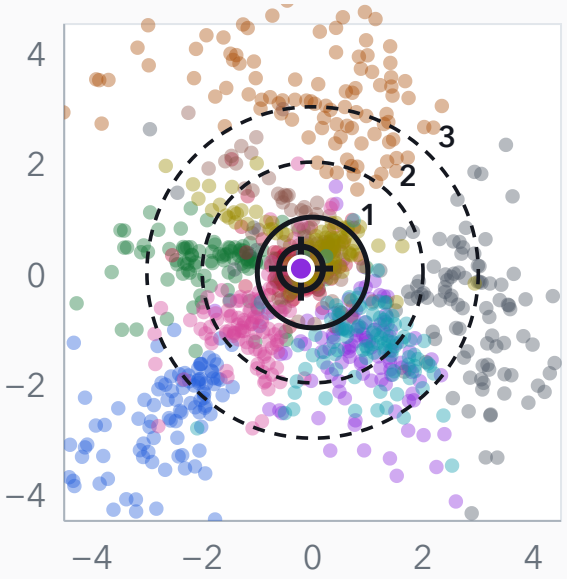


> Walk from a 1 to a 0: every step is a plausible digit. The space between images was trained, because each image owned a cloud.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.7

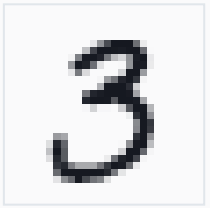
Your turn again: choose a code for the VAE



0 1 2 3 4 5 6 7 8 9

decoder of the VAE, run in your browser





the digit we encoded

circles: N(0, I)
at radius 1, 2, 3

z = (−0.21, 0.07)
the code of our 3, decoded

> Random codes, encoded test digits, or click anywhere. Switch between the VAE and the autoencoder to compare.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook 5.8

Put a weight on the KL: β

$$L_\beta = \mathbb{E}_q\left[\frac{1}{2\sigma^2}\left\|x - f_\theta(z)\right\|^2\right] + \beta\,\mathrm{KL}\left(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I)\right)$$

β = 1 is the true ELBO. Changing β is the same as assuming noisier or cleaner pixels, a different σ.

Small β

codes free to spread: close to an autoencoder

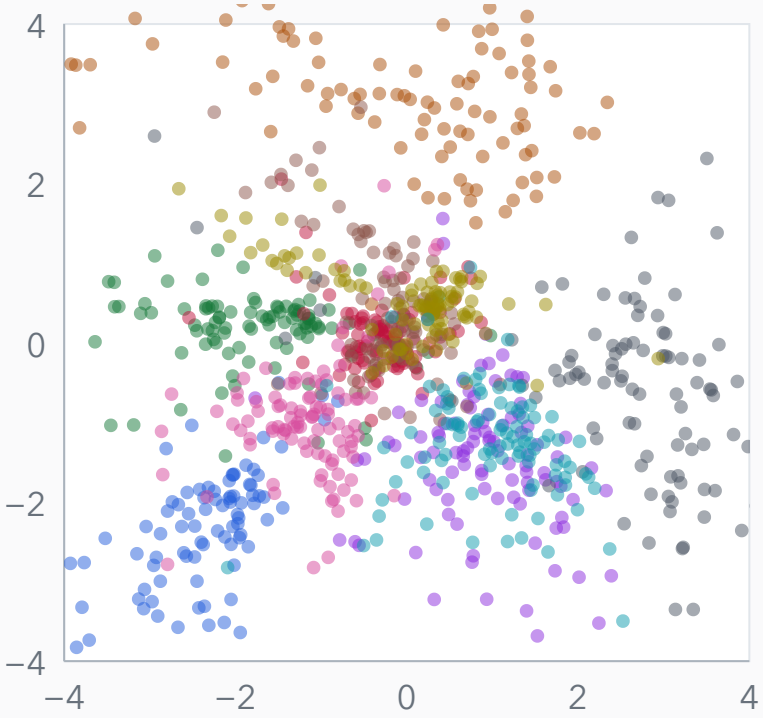
Large β

codes squeezed towards the prior: closer to N(0, I), less detail

> Predict before we look: what happens at β = 0, and at very large β?

▶ Notebook 5.8

Seven VAEs, β from 0 to 256: try them

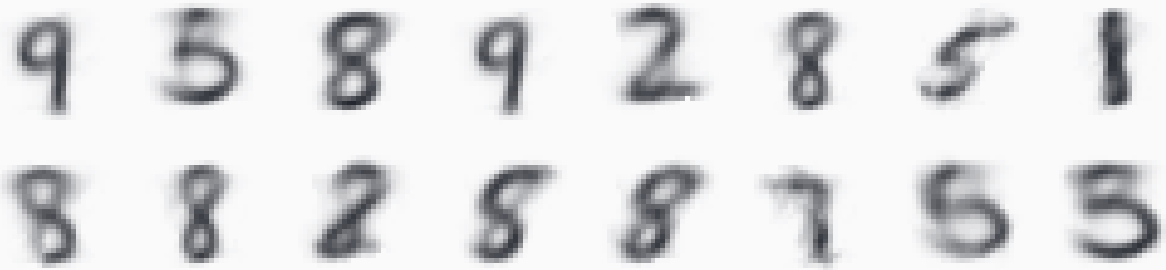


β = 1 KL 9.0 · squared error 30.8 · average σ of q 0.034

reconstructions of 8 test digits



16 samples: z ~ N(0, I), then decode



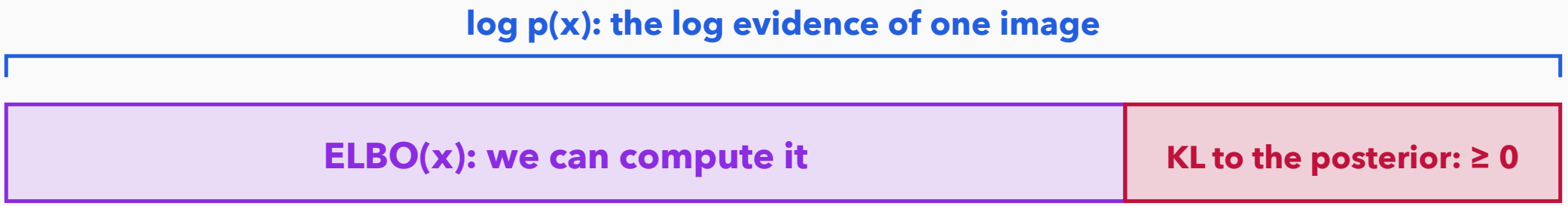
distance between two 0s ÷ distance from a 0 to a 1: 0.23

Click through β. Watch how spread the codes are, the reconstructions, and the samples.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Part 6 · Putting it together



raise the ELBO, and the KL to the posterior must fall

6.1 The whole lecture in maths ▶

6.2 Wrap-up ▶

> Every piece of the lecture on one page: the VAE's loss, term by term, traced back to KL and the ELBO.

▶ Notebook 6.1

The whole lecture in five lines of maths

		the maths	where we met it
①	KL	$\mathrm{KL}(q \,\|\, p) = \mathbb{E}_q\big[\log q - \log p\big] \;\geq\; 0$	Lecture 1, 2.4
②	the budget	$\log p_\theta(x) = \mathrm{ELBO}(x) + \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z \mid x)\big)$	Lecture 1, 4.1 · today 5.4
③	its two terms	$\mathrm{ELBO}(x) = \mathbb{E}_q\big[\log p_\theta(x \mid z)\big] - \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big)$	Lecture 1, 4.1 · today 5.4
④	Gaussian pixels	$-\log p_\theta(x \mid z) = \frac{1}{2\sigma^2}\|x - f_\theta(z)\|^2 + \text{const}$	today 5.2 and 2.3
⑤	the two tools	$z = \mu + \sigma \odot \epsilon,$ $\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, I)\big)$ exactly	Lecture 1, 3.5, 2.6 · today 4.2–4.3

> ⑤ The two tools make the loss computable: draw z by reparameterisation, and compute the KL between two Gaussians exactly.

▶ Notebook 6.1

Where each term of the VAE's loss comes from

reconstruction term

$$\|x - f(\mu + \sigma \odot \varepsilon)\|^2 / 2\sigma^2$$

- Gaussian pixels: it is $-\log p(x \mid z)$ + const — **5.2**
- the autoencoder's squared error, rescaled — **2.3**
- $z = \mu + \sigma \odot \varepsilon$: Fix 2, tool ① — **Lecture 1, 3.5**

+

KL term

$$KL(q(z \mid x) \,\|\, N(0, I))$$

- Fix 3, computed exactly by tool ② — **Lecture 1, 2.6**
- to the prior $p(z) = N(0, I)$: Fix 1 — **4.1**
- a KL: extra cost, never negative — **Lecture 1, 2.4**

together: −ELBO(x) + const, Lecture 1's objective for one image (5.4)

a smaller loss means a higher lower bound on log p(x), and q closer to the posterior

> Together they are minus Lecture 1's ELBO for one image. Making the loss smaller pushes up a lower bound on log p(x).

▶ Notebook 6.2

Autoencoder and VAE, side by side

	autoencoder	VAE
the encoder outputs	a code z: 2 numbers	a cloud: μ and log σ^2, that is $q_\phi(z \mid x)$
the decoder gets	z	a draw $z = \mu + \sigma \odot \varepsilon$ (tool ①)
the loss	squared error	squared error / $2\sigma^2$ + KL to N(0, I) (tool ②) = −ELBO + const
random z ~ N(0, I) gives	mostly junk	digits
can it score an image?	no	yes, through the ELBO, a lower bound on log p(x)

Takeaway. A VAE is an autoencoder trained with Lecture 1's objective. The KL to the prior makes the codes fill N(0, I), so random codes decode to digits.

Both changes come from Lecture 1: the ELBO with a code per image, the reparameterisation trick, and the Gaussian KL.

▶ Notebook 6.2

Next: can we drop the pixel likelihood altogether?

VAE samples are soft because each pixel is an independent Gaussian around a mean image.

Next: a generator trained against a second network that judges real from fake (GANs). Then many small denoising steps (diffusion).

> Lectures 3 and 4 replace it: a learned critic, then repeated denoising.

▶ Notebook

From Lecture 1: one objective we can compute

$$\log p(\mathcal{D}) = \mathrm{ELBO}(q) + \mathrm{KL}\big(q \,\|\, p(\theta \mid \mathcal{D})\big)$$

log p($\mathcal{D}$) is one fixed number. Whatever the ELBO gains, the KL to the posterior loses.

$$\mathrm{ELBO}(q) = \underbrace{\mathbb{E}_{\theta\sim q}\big[\log p(\mathcal{D} \mid \theta)\big]}_{\text{fit the data}} - \underbrace{\mathrm{KL}\big(q(\theta) \,\|\, p(\theta)\big)}_{\text{stay near the prior}}$$

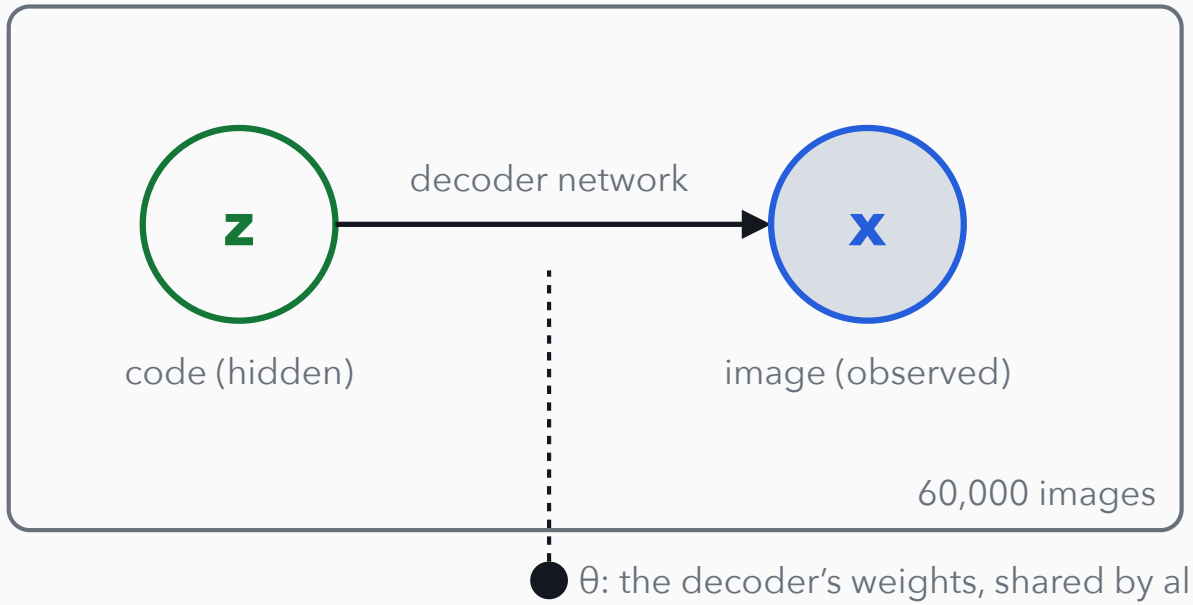
$$\text{train:} \quad \psi^* = \arg\max_{\psi} \ \mathrm{ELBO}(q_\psi) \qquad \text{loss} = -\mathrm{ELBO}$$

↗ Lecture 1, 4.1: the objective
↗ Lecture 1, 4.1: the ELBO in two terms

> The objective: tune q's knobs ψ to maximise the ELBO, by gradient descent on minus the ELBO. Today reuses this line unchanged.

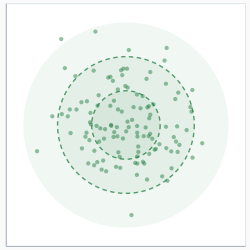
▶ Notebook

The VAE as a graphical model: a hidden code



the prior:

$p(z) = N(0, I)$, fixed in advance



the likelihood:

$p(x \mid z) = N(x; \text{decoder}(z), \sigma^2 I)$

each pixel: the decoder's value + noise of size σ

f(z), then x:



> The decoder's weights θ sit outside the box: one set, shared by every image, learned from the data.

▶ Notebook

Same problem as Lecture 1, new names

	Lecture 1	today, for one image x
the unknown, with a prior	θ: the tiny network's 7 weights	z: this image's 2-number code
prior	$p(\theta)$	$p(z) = \mathcal{N}(0, I)$
likelihood	$p(\mathcal{D} \mid \theta)$	$p_\theta(x \mid z)$: the decoder
posterior: out of reach	$p(\theta \mid \mathcal{D})$	$p(z \mid x)$
approximation	$q_\psi(\theta)$: the guide	$q_\phi(z \mid x)$: the encoder
how many q's?	one	one per image: 60,000, all computed by one encoder network

Careful with θ: today it names the decoder's weights, learned as ordinary parameters. The unknown with a prior is z.

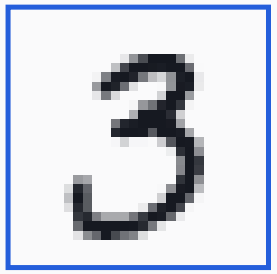
> One change: Lecture 1 fitted one q. Here every image needs its own q, so a network, the encoder, outputs it from the image.

▶ Notebook

The likelihood, written out: Gaussian pixels

$$\log p_\theta(x \mid z) = \sum_{j=1}^{784} \log \mathcal{N}\left(x_j;\ f_\theta(z)_j,\ \sigma^2\right) = -\frac{\|x - f_\theta(z)\|^2}{2\sigma^2} + 541.4$$

the image



x: our 3

in the posterior's patch

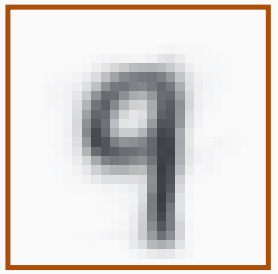


z = (−0.23, 0.06)

squared error
23.1

log p(x | z)
252

a draw from the prior

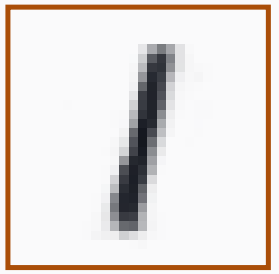


z = (0.82, −0.74)

squared error
72.1

log p(x | z)
−360

where the 1s are



z = (0.00, 3.20)

squared error
83.7

log p(x | z)
−505

the decoder's mean image $f_\theta(z)$ for each code: a code that draws our 3 well gets a high likelihood

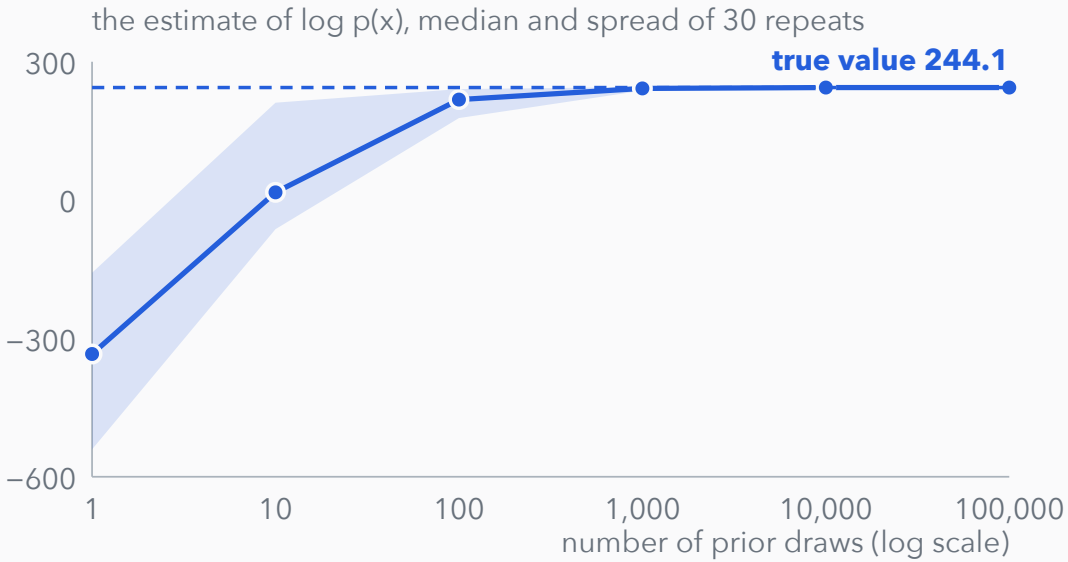
σ = 0.2: about the typical pixel error of a 2-number code, √(31 / 784) ≈ 0.2. The constant is −784 log(σ√2π).

> σ = 0.2 is a modelling choice, matched to a typical pixel error. Smaller σ makes every unit of squared error cost more.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

The evidence p(x): average the likelihood over the prior



$$p_\theta(x) = \mathbb{E}_{z\sim p(z)}\left[p_\theta(x \mid z)\right] \approx \frac{1}{S}\sum_{s=1}^{S} p_\theta(x \mid z_s)$$

Draw S codes from the prior, decode each one, and average the likelihoods.

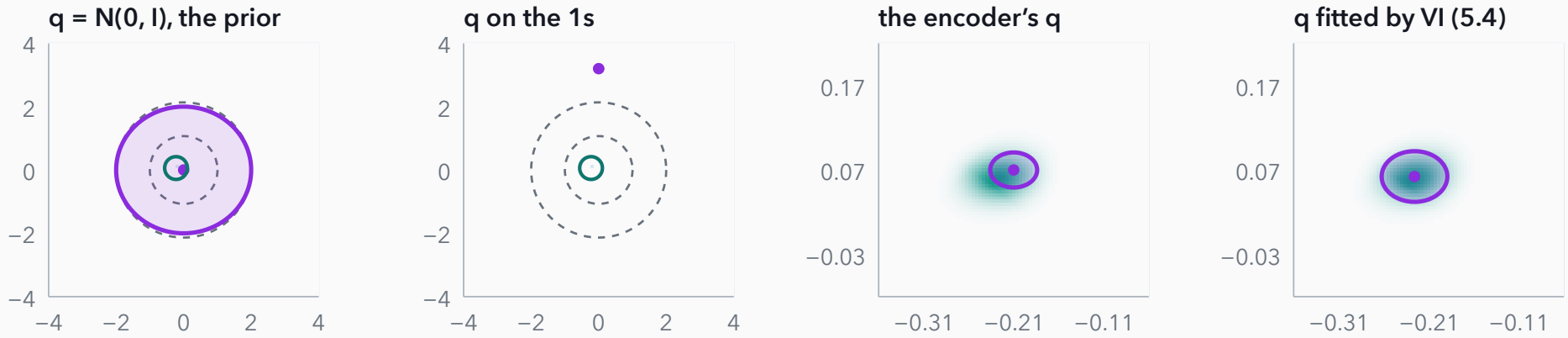
Of 100,000 prior draws, the best 10 carry 38% of the total. Almost every code from the prior explains our 3 terribly.

> Why so many? The codes that explain our 3 form a patch about 0.018 wide, and random codes almost never land in it. For one image in 2 numbers brute force barely works; for 60,000 images, every step, in 16 numbers, it is hopeless.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Put a guide q: a Gaussian guess of where the codes are



ELBO −269.7
gap 514
teal ring: the posterior

ELBO −514.2
gap 758
teal ring: the posterior

ELBO 243.4
gap 0.7
teal: the posterior

ELBO 244.0
gap < 0.1
teal: the posterior

$$\mathrm{ELBO}(q) = \mathbb{E}_{z\sim q}\big[\log p_\theta(x \mid z)\big] - \mathrm{KL}\big(q \,\|\, p(z)\big) \qquad \log p_\theta(x) = \mathrm{ELBO}(q) + \mathrm{KL}\big(q \,\|\, p(z \mid x)\big)$$

> Why this score? log p(x) is one fixed number, and the gap log p(x) − ELBO is exactly the guide's KL to the posterior. The highest ELBO marks the closest guide: gaps 514, 758, 0.7 and < 0.1.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Lecture 1's ELBO, for one image

Lecture 1

$$\mathrm{ELBO}(q) = \mathbb{E}_{\theta \sim q}\big[\log p(\mathcal{D} \mid \theta)\big] - \mathrm{KL}\big(q(\theta) \,\|\, p(\theta)\big)$$

today: swap the names

$$\mathrm{ELBO}(x) = \mathbb{E}_{z \sim q_\phi(z|x)}\big[\log p_\theta(x \mid z)\big] - \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big)$$

$$\log p_\theta(x) = \mathrm{ELBO}(x) + \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z \mid x)\big) \;\geq\; \mathrm{ELBO}(x)$$

Raising the ELBO does two jobs: θ makes the image more probable, and φ brings q closer to the posterior.

> And the same budget: log p(x) = ELBO + a KL that is never negative. Training maximises the ELBO summed over all 60,000 images.

▶ Notebook

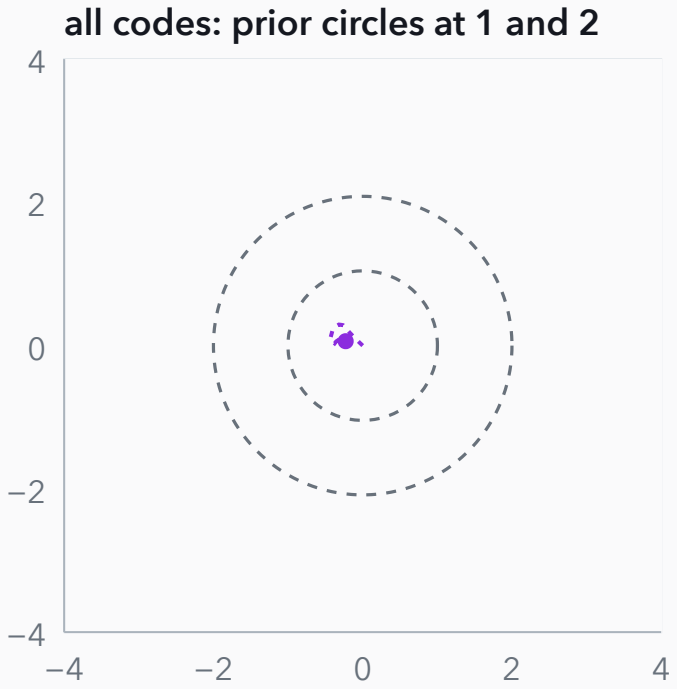
Read the ELBO term by term: it is Part 4's three fixes

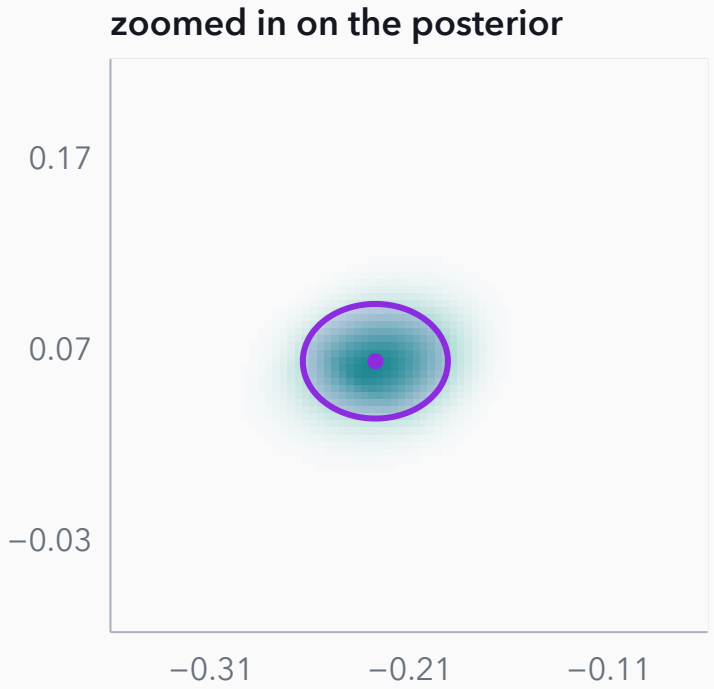
piece of the ELBO for one image	what it says	where we met it
$\mathbb{E}_{z\sim q_\phi(z\|x)}\left[\,\cdots\,\right]$	draw codes from this image's cloud: the cloud is q	Fix 2
$\log p_\theta(x \mid z)$	decode z and score how well it explains x: the rebuild term	the autoencoder's loss (5.2)
$-\mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big)$	pull the cloud towards the prior p(z) = N(0, I)	Fixes 1 and 3

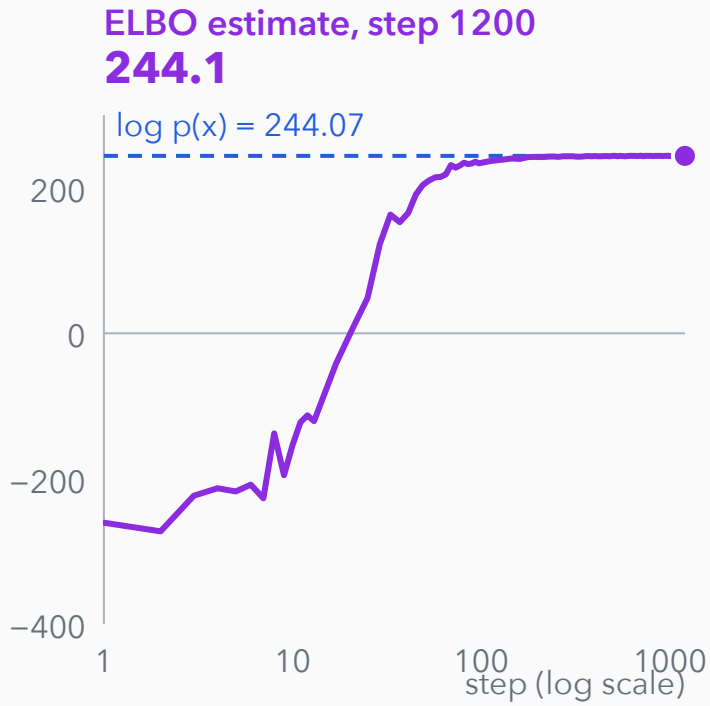
> The KL to the prior is Fix 3's penalty, and the prior N(0, I) is Fix 1. Part 4's guessed loss was minus the ELBO all along.

▶ Notebook

Lecture 1's VI on one image: fit q for our 3, press play







> q moves into the posterior's patch and shrinks to fit it. The ELBO ends at 244.05, just below the ceiling log p(x) = 244.07.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Minus the ELBO is Part 4's loss, with its weight decided

$$-\mathrm{ELBO}(x) = \underbrace{\mathbb{E}_q\left[\tfrac{1}{2\sigma^2}\left\|x - f_\theta(z)\right\|^2\right]}_{\text{from } -\log p_\theta(x|z)} + \underbrace{\mathrm{KL}\big(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I)\big)}_{\text{stay near the prior}} + \underbrace{\text{const}}_{-541.4}$$

$$\mathbb{E}_q\big[\cdots\big] \approx \tfrac{1}{2\sigma^2}\left\|x - f_\theta(\mu + \sigma \odot \epsilon)\right\|^2, \qquad \epsilon \sim \mathcal{N}(0, I) \text{ (one draw, tool ①)}$$

$$\mathrm{KL} = \tfrac{1}{2}\sum_{k=1}^{2}\left(\mu_k^2 + \sigma_k^2 - \log\sigma_k^2 - 1\right) \qquad \text{(tool ②, no sampling)}$$

> The KL between two diagonal Gaussians has a formula, so it needs no sampling. The loss uses both of Lecture 1's tools.

▶ Notebook

Did the autoencoder memorise? Training against test error

2-number code	training error	test error
autoencoder	30.3	31.1
PCA		43.6

Test digits were never used for training.

Training and test errors are close: it learned digits in general, not these particular 60,000.

> The gap is small (30.3 against 31.1), and the autoencoder beats PCA by a wide margin with the same 2 numbers.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Without the ReLUs and the sigmoid, an autoencoder rediscovers PCA



Test error: linear autoencoder 43.8, PCA 43.6.

Its decoder spans almost the same plane as PCA's two directions (cosines of the angles: 1.000 and 0.92).

> Gradient descent found PCA's plane on its own: an autoencoder is PCA with bendable maps.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

The autoencoder: walk in a straight line from a 1 to a 0

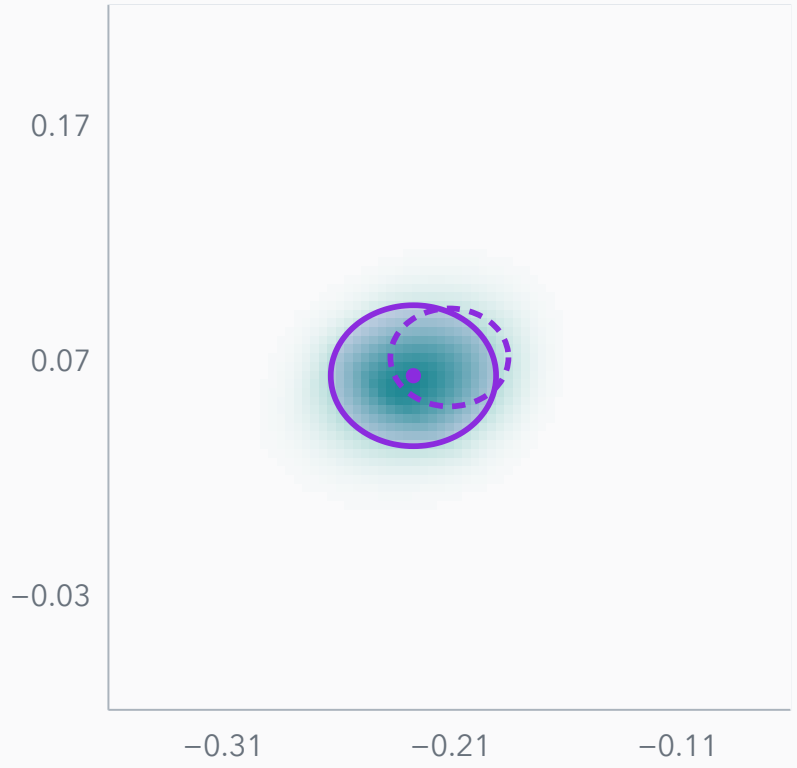


> With 2 numbers the path crosses the clusters of other digits; with 16 the middle passes through blobs that are no digit.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

Check the encoder: its q for our 3, against the true posterior



	value
log p(x), the exact target	244.07
ELBO, q fitted to this image alone	244.05
ELBO, q from the encoder (one pass)	243.37

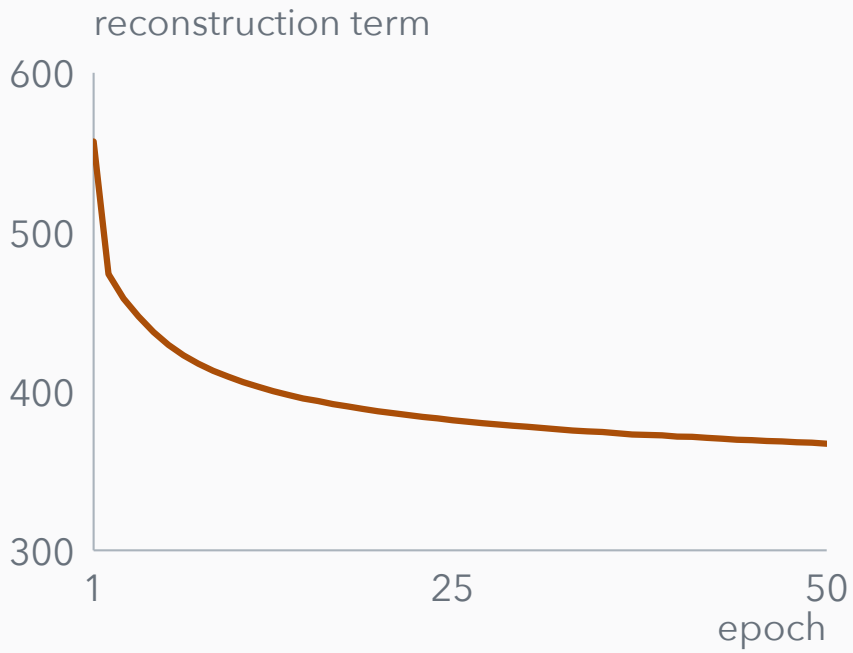
Dashed: the encoder's q. Shaded: the per-image fit. Both 2 standard deviations.

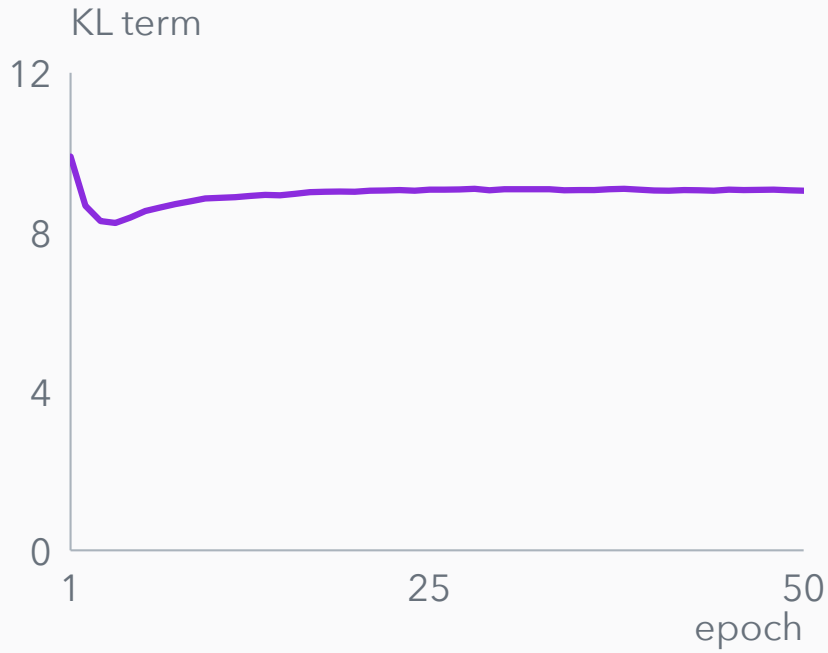
> The encoder's q sits slightly off-centre and a little too narrow: the price of one network serving all 60,000 images.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

While training: reconstruction falls, the KL settles





> After 50 epochs: reconstruction 367, KL 9.0 per image. The KL never falls to zero: then the code would say nothing about the image.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

β = 0: is it just an autoencoder?

β	KL per image	average σ of q	spread of codes
0	68.3	0.0001	5.36
1	9.0	0.0343	1.68
256	0.0	1.0001	0.00

With β = 0 nothing stops q from shrinking.

It still draws $z = \mu + \sigma\varepsilon$, unlike an autoencoder. But training drives σ to 0.0001, so it behaves like one.

> But training drives σ almost to zero, and then it behaves like one: spread-out codes and poor samples.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).

▶ Notebook

β = 256: every code decodes to the same blur

β = 256, rebuilt



β = 256, samples



KL = 0.00: q(z | x) equals the prior for every image.

The decoder learns to ignore z and outputs the average digit. This is called **posterior collapse**.

> Then z carries no information, so the decoder ignores it and draws the average digit every time.

Networks and numbers from Lecture 2: `lecture2/scripts/train.py` and `science.py` (MNIST, fixed seeds).